

SCHOOL OF INFORMATION SCIENCES AND TECHNOLOGY

BYZANTINE FAULT-TOLERANT MULTI-AGENT SYSTEMS FOR AUTONOMOUS RUNTIME PROGRAM REPAIR IN E-COMMERCE SYSTEMS

A PROJECT REPORT SUBMITTED IN PARTIAL FULFILMENT OF THE
REQUIREMENTS FOR THE AWARD OF THE DEGREE OF
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BY

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MAY 2026

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DECLARATION BY THE CANDIDATE

I, Millicent Mufambi (H240624A), hereby declare that the project report entitled “Byzantine Fault-Tolerant Multi-Agent Systems for Autonomous Runtime Program Repair in E-Commerce Systems” carried out by me under the guidance of Mr W. Makondo is submitted in partial fulfilment of the requirements for the award of the degree of Master of Technology in Software Engineering. This is a record of bona-fide work carried out by me and the results embodied in this project have not been reproduced/copied from any source.

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ABSTRACT

When software breaks in a live e-commerce system, the bill arrives quickly, and a hand-written patch rarely ships fast enough to stop the bleeding. Multi-agent large-language-model (LLM) systems can now write plausible patches on their own, yet they talk their way to a decision and set no limit on how many agents may be confidently wrong before a bad fix is shipped. This project builds and tests BFT-MAS, a Byzantine fault-tolerant multi-agent system that adapts the Practical Byzantine Fault Tolerance (PBFT) protocol to a mixed group of LLM repair agents, backed by an isolated execution sandbox and e-commerce integrity checks. A Claude analyzer and a GPT-4o healer propose a fix that four independent, model-diverse validators (Claude-Haiku, GPT-4o-mini, llama3.1:8b and mistral:7b) then vote on under PBFT ($n = 3f + 1 = 4$, quorum = 3) before a sandbox stage, measured against a single-agent baseline on forty synthetic Python bugs, twenty real bugs from BugsInPy, and a thirty-bug e-commerce suite. The study demonstrates the core Byzantine property directly: on an independent-replica realisation, a malicious primary that equivocates makes a naive vote split the honest replicas onto different fixes, while the PBFT quorum certificates preserve agreement and reject forged messages; fault injection and an $f = 2$ study further show the quorum masks faulty votes up to the $3f + 1$ bound. Repair results are reported as preliminary: on the real-world subset the unsafe-fix rate fell directionally from 75% to 50% and the repair rate rose from 25% to 50%, but neither reaches significance (McNemar exact $p = 0.125$) and the effect rests on a weak run-as-script oracle. The work delivers an open-source prototype, a repeatable harness, and the first empirical study of genuine BFT consensus applied to LLM-driven repair in e-commerce.

Keywords: Byzantine fault tolerance, multi-agent systems, large language models, automated program repair, PBFT consensus, e-commerce reliability, software engineering.

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CHAPTER 1: INTRODUCTION AND BACKGROUND TO THE STUDY

1.1 Introduction

Almost everything about modern commerce now runs on software. Retail front-ends, payment gateways, inventory services and order-management systems stay up around the clock, and the correctness of their code feeds straight into customer trust, revenue and reputation. Let a defect slip into production and the damage shows up at once: a shopper is charged twice, a stock count slides below zero, a refund sails past its cap, or an order is confirmed while the payment is still pending. The Consortium for Information and Software Quality (CISQ, 2024) puts the cost of critical production failures at roughly one million United States dollars an hour for large firms, and as much as five million an hour for busy retailers during peak trading. How quickly, and how safely, a defect can be found and fixed is therefore not a back-office nicety; for an online business it is close to a survival metric.

For most of software engineering's history, that find-and-fix loop has been human work. Someone is paged, reproduces the failure, reasons out the cause, writes a patch, tests it and ships it. Every one of those steps costs time, and in e-commerce time is money. Automated program repair (APR) grew up to shorten the loop by proposing fixes automatically. The first APR tools searched over program mutations or solved constraints to synthesise a patch; newer ones lean on the code-writing ability of large language models (LLMs), which produce fixes that are often plausible and sometimes right. Alongside this, capable LLMs have spawned multi-agent systems, where several model-backed agents split a software task between them, one analysing, one writing, one reviewing, in a rough imitation of a human team.

This project lives where three of those developments meet: Byzantine fault tolerance, multi-agent LLM systems and automated program repair. The motivation is one stubborn observation. Today's multi-agent LLM frameworks let their agents talk their way to a decision. There is no formal idea of agreement, and nothing caps how many agents can be confidently wrong before the group ships a bad fix. When one bad patch can scramble a payment ledger or oversell stock, that missing guarantee is a real problem. Distributed-systems researchers cracked a very similar problem decades ago with Byzantine fault tolerance, a family of protocols that still reach a correct collective decision even when some bounded number of participants misbehave in arbitrary ways. The question this project takes up is whether that guarantee can be carried over to a population of non-deterministic LLM agents doing repair,

and what is won and lost in the process. The rest of this chapter sets out the background, states the problem and why it matters, lists the objectives and research questions, fixes the scope, defines the key terms, and ends with a map of the report.

1.2 Background

The theoretical bedrock here is the Byzantine Generals Problem, set out by Lamport, Shostak and Pease in 1982. They showed that to reach agreement when f participants may fail arbitrarily, including by feeding different lies to different peers, a system needs at least $n = 3f + 1$ participants in total. Tolerating one arbitrarily faulty participant thus takes four; tolerating two takes seven; and so on. That $3f + 1$ floor is the skeleton of every fault-tolerant consensus protocol since. A second classic result tempers the first: the impossibility theorem of Fischer, Lynch and Paterson (1985) proves that in a fully asynchronous network, no deterministic protocol can promise both safety and progress in the face of even a single crash. This is why every practical protocol leans on some extra assumption about timing, randomness or trusted hardware.

Castro and Liskov (1999) turned the theory into something usable with the Practical Byzantine Fault Tolerance protocol, or PBFT. It agrees in three message rounds, pre-prepare, prepare and commit, and was the first protocol lean enough to run a real replicated service, adding only a few per cent of overhead to a networked file system. PBFT is still the reference point, and later protocols, Zyzzyva, BFT-SMaRt, RBFT, Aardvark, HotStuff and SBFT among them, mostly sharpen its speed or its resilience. All of them, though, lean on one assumption: the replicas are deterministic. Feed two correct replicas the same input and they return byte-for-byte the same output, so any disagreement is, by itself, proof of a fault. That is what makes a vote mean something in the classical world.

The second thread is the fast rise of multi-agent LLM systems. Frameworks like ChatDev (Qian et al., 2024), MetaGPT (Hong et al., 2024), AutoGen (Wu et al., 2024) and CAMEL (Li et al., 2023) wire several LLM-backed agents into teams that chase software tasks by passing natural-language messages around. They are genuinely capable, but their coordination is social, not formal: roles, peer review and discussion stand in for any guaranteed agreement rule. And there is a catch. LLM agents are stochastic; they sample their outputs with a dash of randomness, so two equally good agents can hand back different yet equivalent fixes for the same bug. That single fact breaks the old equation of correctness with identical output, and it makes a straight transfer of Byzantine consensus to the LLM world far from obvious.

The third thread is automated program repair. It moved from search-based mutation, GenProg (Le Goues et al., 2012) being the classic case, through constraint-based synthesis such as SemFix (Nguyen et al., 2013) and Angelix (Mechtaev et al., 2016), to LLM-driven methods that treat a fix as code completion (Chen et al., 2021; Xia and Zhang, 2022). One weakness shadows the whole field: the plausibility-correctness gap that Smith et al. (2015) pinned down, where a patch passes the available tests yet is still wrong. In their numbers, just two of fifty-five generated patches that passed the failing test held up on held-out tests. That precise failure, a confident patch that is quietly incorrect, is exactly what a diverse group of agents voting together, with an independent execution check behind them, ought to be able to catch. The meeting of these three threads, the safety of Byzantine consensus, the fluency of multi-agent LLMs and the verification headache of repair, marks out the space this project works in.

1.3 Problem Statement

LLM-driven repair has become good enough that one can seriously imagine letting it fix defects in live e-commerce systems with little human oversight. The trouble is that the multi-agent frameworks driving these models hand back no formal safety guarantee. They reach decisions by conversation, set no limit on how many agents may be confidently wrong, never report their own false-positive rate, and do nothing to protect against an agent that goes rogue, whether through a model failure, a prompt-injection attack or a plain hallucination. In a domain this unforgiving, one confident but wrong fix, accepted and deployed, can break a payment, inventory or refund invariant and cost real money.

Distributed systems already has a mature answer to arbitrary failure in Byzantine fault-tolerant consensus, but every classical protocol assumes deterministic replicas and has never been adapted to, let alone tested on, a population of non-deterministic LLM agents. So there is no published system that brings formal consensus to LLM-driven repair, no evidence on whether such consensus actually cuts unsafe fixes, no evaluation aimed at e-commerce, and no measurement of whether a mixed agent population really produces the decorrelated failures that consensus banks on. The gap this project fills is the lack of a formally grounded, empirically tested way to keep multi-agent LLM repair safe in production e-commerce systems.

1.4 Justification

The case for this work is partly economic and partly scientific. On the economics, the cost of unsafe automated repair in e-commerce is lopsided: a fix that is quick but wrong can hurt far

more than one that is a little slower but right, because a bad change to payment or inventory logic can corrupt financial state without anyone noticing. So a mechanism that buys the elimination of unsafe fixes for a modest latency increase pays for itself wherever each repair is a high-stakes commit. On the CISQ (2024) figures, even a small drop in the rate of unsafe fixes reaching production adds up to large avoided losses across many high-value repairs.

On the science, the project closes a specific gap between two mature literatures that had barely spoken to each other. As far as I am aware, it is the first study to run a classical PBFT protocol over a mixed population of LLM agents, to report an empirical safety-violation rate for such a system, to bake e-commerce-specific invariants into the repair loop, and to measure agent-diversity decorrelation directly instead of taking it on faith. A working prototype, a repeatable benchmark and a statistical comparison against a single-agent baseline give the field hard evidence where there had been only architectural argument. Reporting the costs as plainly as the benefits, the doubled latency and the negative decorrelation result alongside the elimination of unsafe fixes, sharpens the contribution by saying exactly when the approach earns its keep and when it does not.

1.5 Objectives

The broad aim of the project is to design, build and empirically validate a Byzantine fault-tolerant multi-agent system for autonomous runtime program repair in e-commerce systems. That aim breaks into five specific objectives:

1. **To** design and implement a multi-agent architecture comprising an analyzer, a healer and validator agents, each backed by a different LLM provider, and coordinate them with a three-phase PBFT consensus engine;
2. **To** embed an isolated execution sandbox and a set of e-commerce-specific integrity invariants (covering payment idempotency, non-negative inventory, refund caps and currency coherence) as a final correctness check on every consensus-approved fix;
3. **To** evaluate the system empirically against a single-agent baseline across a synthetic benchmark, a real-world bug subset and a purpose-built e-commerce bug suite, using a defined set of consensus and safety metrics;
4. **To** quantify the failure-decorrelation property of a heterogeneous validator population by measuring pairwise validator agreement, and to test the system's Byzantine fault-tolerance property under injected adversarial agent behaviours; and

5. To release the prototype, evaluation harness and datasets as an open-source contribution to enable reproduction and extension by other researchers.

1.6 Research Questions

In support of these objectives, the study is guided by four research questions:

1. **RQ**What is the optimal number of agents required to achieve Byzantine fault tolerance without incurring excessive latency?
2. **RQ**How does consensus latency affect real-time repair performance?
3. **RQ**Can multi-agent consensus improve repair reliability compared with single-agent systems?
4. **RQ**What economic benefits can e-commerce businesses derive from the reduced downtime that safer automated repair makes possible?

1.7 Scope and Significance of the Project

The scope is drawn tightly on purpose, so that every claim can be backed by evidence. The system targets Python, because the validators' static checks are language-specific; carrying the design to other languages such as Java is left for later. Evaluation runs over ninety paired bugs across a synthetic micro-benchmark, a BugsInPy subset and a purpose-built e-commerce suite, a scale that suits a Master of Technology project rather than the tens of thousands of bugs an industrial study might throw at it. Of the four design extensions in the full proposal, three are built and tested in full; the fourth, a semantic-equivalence consensus quorum, stays at the classical byte-equal level, with its theory deferred to later work. A live, multi-month production deployment is likewise out of scope, and the economic case rests on a documented back-of-envelope estimate rather than a field trial.

Inside those limits the project still matters a good deal. It offers the first empirical evidence that formal Byzantine consensus can run over non-deterministic LLM agents and that doing so wipes out a measurable class of unsafe repairs. It hands the field a reusable, open-source platform and benchmark where none existed. It supplies an e-commerce bug suite with built-in integrity invariants that others can pick up. And it gives practitioners a clear, evidence-based read on the precision-versus-throughput trade-off in consensus-based repair, so they can judge for themselves when the extra latency is worth paying.

1.8 Definition of Key Terms

Byzantine fault. An arbitrary fault in which a component may behave in any manner, including sending different and conflicting information to different participants, rather than simply crashing.

Byzantine fault tolerance (BFT). The property of a distributed system that allows it to reach a correct collective decision despite a bounded number of Byzantine faults among its participants.

Practical Byzantine Fault Tolerance (PBFT). The consensus protocol of Castro and Liskov (1999) that achieves Byzantine agreement through a three-phase (pre-prepare, prepare, commit) message exchange with a quorum of $2f + 1$ out of $3f + 1$ participants.

Large language model (LLM). A neural network trained on very large text and code corpora that generates text or code by predicting likely continuations of a prompt.

Agent. In this project, an autonomous software component backed by an LLM that performs a defined role (analysis, repair or validation) within the repair pipeline.

Automated program repair (APR). The automatic generation of source-code changes that fix a defect so that the program satisfies its specification or passes its tests.

Safety violation. In this project, a fix that is approved by the system but subsequently fails execution in the isolated sandbox, indicating that an incorrect change would have been committed.

Sandbox. An isolated execution environment in which a candidate fix is run without affecting the original codebase, used as a final correctness check.

Failure decorrelation. The degree to which the errors made by different agents are statistically independent; high decorrelation is what allows a vote among diverse agents to outperform any single agent.

1.9 Conclusion

This chapter has placed the study at the junction of Byzantine fault tolerance, multi-agent LLM systems and automated program repair. The argument so far is straightforward: LLM-driven repair is now capable enough to belong in production e-commerce systems, yet it lacks the formal safety guarantee that such a setting demands, and that missing guarantee is the heart of

the research problem. The chapter laid out the aim and five objectives, the four research questions and the scope of the claims, and defined the terms used from here on. Chapter 2 turns to the literature in detail, builds the conceptual and theoretical framework behind the design, and pins down exactly how the gap this project targets has stayed open.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter reviews the knowledge the project stands on and pins down the gap it fills. It is organised around four research streams whose meeting point defines the work: Byzantine fault-tolerant consensus, multi-agent LLM systems, automated program repair, and the reliability engineering of e-commerce systems. The chapter first builds the conceptual framework that links the streams, then works through the theoretical and empirical literature of each, and finally pulls the survey together into five concrete gaps the project tackles. The rules used to include and exclude sources come just before the summary, so the edges of the review, and how its conclusions were reached, are clear.

2.2 Conceptual Framework

The conceptual framework turns on one analogy and one tension. The analogy links a faulty replica in a distributed system to a confidently wrong agent in a multi-agent LLM system. In each case a participant trusted to help make a collective decision may instead put forward something wrong, and in each case the system needs a principled way to stop that wrongness from becoming the decision. Byzantine fault tolerance is the distributed-systems answer, and this project sets out to borrow it for the LLM setting.

The tension sits between what classical consensus assumes and what LLM agents actually are. Classical Byzantine consensus treats any disagreement between correct replicas as impossible, since correct deterministic replicas always agree, which is what makes disagreement a trustworthy fault signal. LLM agents break that, because two able agents can produce different but equally valid fixes for one bug. The framework escapes the tension by splitting agreement in two: agreement on a decision, whether to apply a fix at all, which ordinary voting can settle, and agreement on a specific artefact, the exact text of the fix, which in the LLM world calls for semantic equivalence rather than byte equality. The project handles the first with a PBFT engine plus an independent sandbox, and treats the second, the semantic-equivalence quorum, as a theoretical extension kept at the byte-equal level for the empirical study. Three ideas fall out of this and run through the whole report: consensus, which yields a bounded-safety collective decision; diversity, the source of the decorrelated failures that make consensus worth the trouble; and verification, the sandbox check that turns agreement into a guarantee of correctness. Figure 2.1 sketches how the three fit together, and Chapter 3 makes them concrete.

Conceptual Framework: Three Disjoint Research Traditions
and the Intersection Targeted by This Review

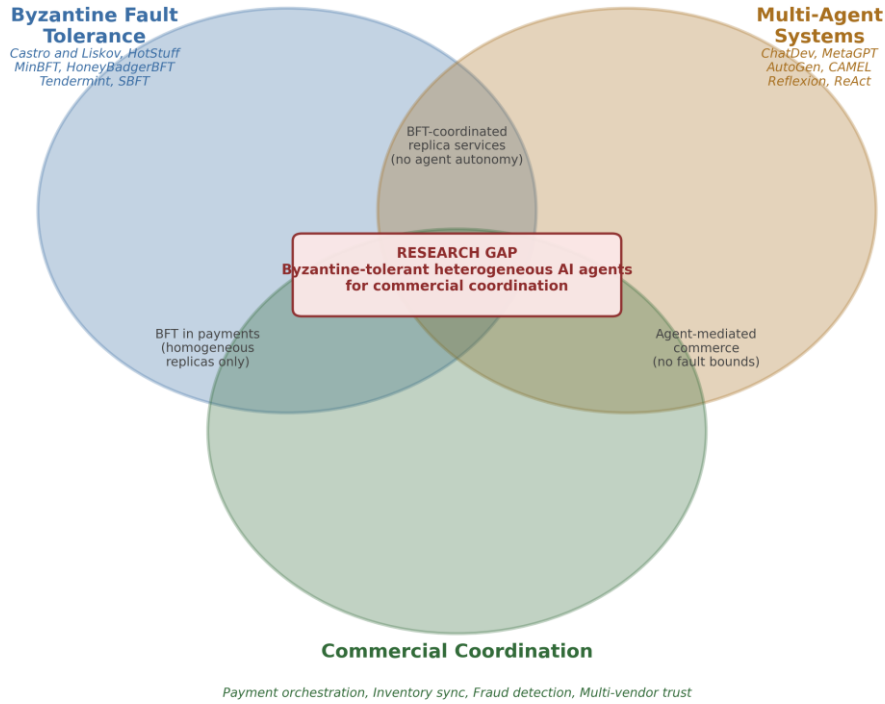


Figure 2.1. Conceptual framework relating consensus, diversity and verification in the proposed system.

2.3 Theoretical and Empirical Literature

2.3.1 Byzantine Fault Tolerance: From Theory to Practice

Getting independent participants to agree when some of them may be faulty is one of the oldest problems in distributed computing. Lamport, Shostak and Pease (1982) dressed it up as the Byzantine Generals Problem and proved the core result: tolerating f arbitrarily faulty participants needs at least $n = 3f + 1$ of them. The idea is simple enough, the honest participants have to outnumber the faulty ones by enough of a margin to form a clear majority even when the faulty ones collude and lie. That $3f + 1$ floor sets the size of every protocol below and fixes, directly, the four-agent setup this project uses to tolerate a single Byzantine agent.

Fischer, Lynch and Paterson (1985) added a sobering companion result, the FLP impossibility theorem. In a fully asynchronous system, where messages may be delayed for any length of time, no deterministic protocol can promise to stay both safe and live even if just one participant

might crash. Real protocols dodge this by giving up one of the assumptions: assuming partial synchrony, where messages do eventually arrive within some unknown bound (Dwork, Lynch and Stockmeyer, 1988); adding randomisation (Miller et al., 2016); or leaning on trusted hardware (Veronese et al., 2013). FLP is the reason the consensus engine here, like every practical engine, runs under a partial-synchrony timing assumption.

With PBFT, Castro and Liskov (1999) made Byzantine fault tolerance practical. Agreement happens over three message phases. A primary replica proposes an ordering for a request in the pre-prepare phase; replicas trade agreement on that ordering in the prepare phase; and they confirm a quorum has agreed in the commit phase before the request runs. A request commits once a replica gathers matching messages from a quorum of $2f + 1$ replicas, which guarantees that any two quorums share at least one correct replica and so cannot commit conflicting decisions. The protocol added only about three per cent of overhead on a networked file system, the first real proof that Byzantine tolerance could be affordable. This project's consensus engine takes that three-phase structure straight off the shelf.

Plenty of later work has sharpened PBFT. Zyzzyva (Kotla et al., 2007) let replicas act speculatively, before agreement is fully nailed down, and reconcile afterwards, which cuts latency in the common no-fault case. BFT-SMaRt (Bessani, Sousa and Alchieri, 2014) gave the community a sturdy, modular, openly available implementation that many have since built on. RBFT (Aublin, Mokhtar and Quema, 2013) ran several protocol instances side by side to expose and contain a faulty primary's damage, while Aardvark (Clement et al., 2009) was built to degrade gracefully under sustained attack rather than only when conditions are benign. A newer wave cut communication cost: HotStuff (Yin et al., 2019) reached linear message complexity per decision with pipelined, threshold-signature phases, and now sits under several blockchains, and SBFT (Gueta et al., 2019) mixed collectors and fast paths to scale to larger groups. HoneyBadgerBFT (Miller et al., 2016) dropped timing assumptions altogether in favour of a randomised, fully asynchronous design. Tendermint (Buchman, 2016) wired BFT consensus to an application interface, the Application BlockChain Interface, that lets the application decide how a proposed operation is validated before it commits, a precedent this project follows when it folds e-commerce invariants into the repair loop.

Two properties hold across this whole lineage and both matter here. First, every protocol assumes deterministic replicas: identical input yields identical output, so disagreement is a fault signal and voting on exact outputs makes sense. Second, every protocol tolerates at most f

faulty replicas out of $3f + 1$. Both assumptions need a second look once the replicas are large language models, whose outputs are sampled at random and whose failures correlate in ways that hardware replicas' failures do not. Re-examining them is one of the project's central contributions.

2.3.2 Multi-Agent Large Language Model Frameworks

The second stream is about herding large language models into teams of agents. ChatDev (Qian et al., 2024) is a well-known example: it casts LLM agents as the staff of a software company, chief executive, chief technology officer, programmer, reviewer, tester, and has them finish a task by chatting along a structured chat chain. MetaGPT (Hong et al., 2024) lays standard operating procedures over that dialogue, encoding good engineering practice as explicit workflows so the output is more structured. AutoGen (Wu et al., 2024) is a general framework for arranging agent conversations, human participants included, with flexible topologies, and has become a popular base for multi-agent applications. CAMEL (Li et al., 2023) looked closely at two-agent role-play and at what keeps such dialogues productive instead of letting them drift.

Related work has made individual agents and small ensembles more reliable. Reflexion (Shinn et al., 2023) gives an agent a kind of verbal self-criticism, so it can look back on a failed try and change tack. ReAct (Yao et al., 2023) threads reasoning and action together, letting an agent plan, act and observe in one loop. Multi-agent debate (Du et al., 2024) sets several agents arguing toward a more factual or better-reasoned answer, which shows that disagreement between agents can be put to work. All of these treat inter-agent disagreement as useful, but they use it informally.

The common thread across this lineage is that coordination runs on social machinery, role authority, peer critique, agreement hashed out in conversation, rather than a formal consensus protocol with provable guarantees. Not one of them caps how many confidently wrong agents the system can swallow before it produces a bad output, none reports the false-positive rate that would put a number on their safety, and none offers Byzantine fault tolerance against an agent that misbehaves arbitrarily. They give coordination capability, not coordination safety. The second mismatch cuts deeper. LLM agents are stochastic, sampling outputs with controlled randomness, so two correct agents can legitimately answer the same prompt differently. Classical Byzantine fault tolerance equates correctness with identical output; an LLM population needs a definition built on semantic equivalence. Closing that gap with a properly

justified semantic-equivalence quorum is an open problem, and this project keeps it at the classical byte-equal level and names it as the main theoretical follow-up.

2.3.3 Automated Program Repair

Automated program repair is the proving ground for the project's design. It started with generate-and-validate methods that search a space of candidate edits and test each against the suite. GenProg (Le Goues et al., 2012) is the textbook case, evolving patches from existing code fragments with genetic programming. Constraint-based methods came next, synthesising repairs that provably meet a specification pulled from the failing and passing tests: SemFix (Nguyen et al., 2013) cast repair as constraint solving over one suspicious statement, and Angelix (Mechtaev, Yi and Roychoudhury, 2016) stretched the idea to multi-line repairs with a lightweight symbolic account of the wanted behaviour. Learning-based methods followed, with DeepFix (Gupta et al., 2017) using a neural network to fix C compiler errors in student code and TFix (Berabi et al., 2021) handling repair as a text-to-text job on a transformer.

Code-trained large language models changed the field once more. Chen et al. (2021) evaluated Codex and showed real program-synthesis muscle, while AlphaRepair (Xia and Zhang, 2022) demonstrated that a zero-shot, cloze-style use of a code model could mend real bugs with no task-specific training. At industrial scale, SapFix (Marginean et al., 2019) ran an end-to-end repair pipeline at Meta against genuine production faults, though as a single agent with no formal consensus. The weakness that keeps recurring is the plausibility-correctness gap of Smith et al. (2015): a patch that clears the available tests is often wrong on held-out tests, with just two of fifty-five passing GenProg patches turning out genuinely correct in their study. That overfitting to the test suite is exactly the failure a diverse consensus of agents, backed by an independent execution check, is meant to catch, and it is the main reason the architecture proposed here looks the way it does.

Two benchmarks anchor repair evaluation. Defects4J (Just, Jalali and Ernst, 2014) curates reproducible real bugs from open-source Java projects, and BugsInPy (Widyasari et al., 2020) does the same for Python, pulling real bugs with their fixes and tests from well-known open-source projects. Since the validator stack here runs Python-specific static checks, a subset of BugsInPy carries the real-world part of the evaluation, and Defects4J is flagged for future cross-language work.

2.3.4 Reliability and Integrity of E-Commerce Systems

The fourth stream is the setting the system is meant for. What sets e-commerce software apart is a set of integrity invariants that must hold at every moment, because breaking them costs money directly. Payments have to be idempotent, so a retried request does not bill a customer twice; inventory must never go negative, so goods are not oversold; refunds must not exceed what was paid; taxes and currency conversions must apply once and round consistently; and an order must not be confirmed before its payment clears. The price of breaking these is well documented. Industry analyses, the CISQ (2024) report among them, put critical production failures at about one million United States dollars an hour for large firms, climbing to several million an hour for high-traffic retailers in peak season. Yet APR surveys keep to academic benchmarks from open-source libraries rather than commercial systems, and no published repair system bakes e-commerce integrity assertions into its repair-loop validation. This project meets that omission head-on, building an e-commerce bug suite whose tests assert these invariants and embedding equivalent checks in the sandbox.

2.3.5 Synthesis: Five Gaps at the Intersection

Lining the four streams up next to one another exposes five concrete gaps where they intersect, and the project takes on each. They are stated here and gathered in Table 2.1 and Table 2.2.

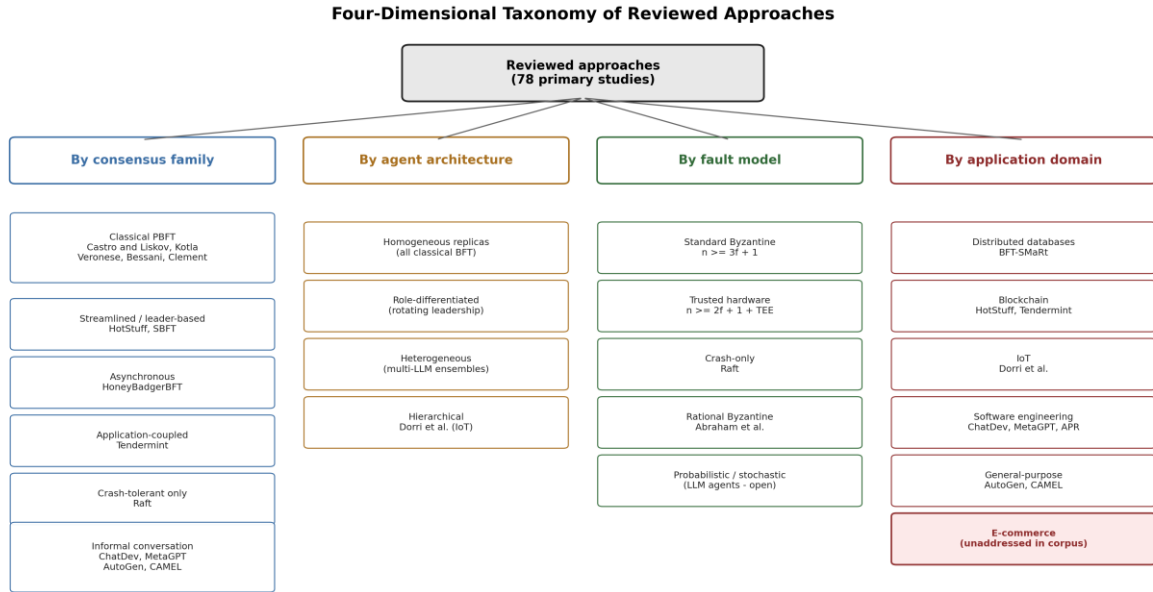


Figure 2.2. Taxonomy of related approaches across the Byzantine-consensus, multi-agent-LLM and automated-program-repair literatures.

- 1. Gap** Existing multi-agent LLM frameworks coordinate through conversation and run no formal agreement protocol; none bounds the number of confidently incorrect agents it can tolerate.
- 2. Gap** No Byzantine fault-tolerant protocol has been adapted to non-deterministic agents; every classical protocol assumes deterministic replicas, and no published rule based on semantic equivalence carries a formal proof.
- 3. Gap** There is no empirical safety evaluation of LLM-agent coordination; the multi-agent literature reports task-completion metrics, and the classical BFT literature reports throughput and latency on deterministic machines where false positives cannot arise.
- 4. Gap** The e-commerce domain is unaddressed; no repair system embeds e-commerce integrity invariants into its validation loop.
- 5. Gap** The failure-decorrelation property of agent diversity is assumed but never measured; no study publishes a pairwise-agreement metric for a heterogeneous LLM population.

Table 2.1. *Gap analysis at the Byzantine-fault-tolerance and multi-agent-LLM intersection.*

System / Family	Formal consensus	Tolerates non-determinism	Empirical safety eval.	E-commerce focus	Diversity measured
PBFT (Castro & Liskov, 1999)	Yes	No	Throughput only	No	N/A
BFT-SMaRt (Bessani et al., 2014)	Yes	No	Latency only	No	N/A
HotStuff (Yin et al., 2019)	Yes	No	Throughput only	No	N/A
HoneyBadgerBFT (Miller et al., 2016)	Yes	No	Latency only	No	N/A
ChatDev (Qian et al., 2024)	No	Yes	Task completion	No	No
MetaGPT (Hong et al., 2024)	Partial (voting)	Yes	Task completion	No	No
AutoGen (Wu et al., 2024)	No	Yes	None	No	No
Multi-agent debate (Du et al., 2024)	No	Yes	Factuality only	No	Implicit only
SapFix (Marginean et al., 2019)	No	Partial	Test-time	No	No
BFT-MAS (this project)	Yes (PBFT)	Yes (sandbox quorum)	BugsInPy 20 pairs, safety 75% to 50% (McNemar $p = 0.125$, n.s.); BFT proven by the fault matrix and the equivocating-primary test	Yes (30-bug suite)	Yes (measured)

Table 2.1 makes the gap visible at a glance. The classical BFT systems in the upper rows provide formal consensus but assume determinism, report only performance metrics and have no notion of an e-commerce domain. The multi-agent and repair systems in the middle rows tolerate non-determinism and perform real tasks but provide no formal consensus and no safety

evaluation. Only the system proposed in this project, in the final row, occupies the intersection of all five columns.

Table 2.2. *Feature comparison of representative automated-program-repair approaches.*

System	Multi-agent	Byzantine consensus	Sandbox check	Domain invariants	Code generation
GenProg (Le Goues et al., 2012)	No	No	No	No	Mutation
SemFix (Nguyen et al., 2013)	No	No	No	No	Synthesis
DeepFix (Gupta et al., 2017)	No	No	No	No	Neural
TFix (Berabi et al., 2021)	No	No	No	No	Transformer
SapFix (Marginean et al., 2019)	No	No	Partial (test-time)	No	Template + ML
ChatDev (Qian et al., 2024)	Yes	No	No	No	LLM
MetaGPT (Hong et al., 2024)	Yes	Partial (voting)	No	No	LLM
BFT-MAS (this project)	Yes	Yes (PBFT)	Yes (isolated)	Yes (e-commerce)	LLM

2.4 Inclusion and Exclusion Criteria

Since the review reaches across four streams, explicit rules kept it focused and repeatable. A source was included if it cleared a few bars. It had to be a peer-reviewed paper, an established preprint or an authoritative technical report from between 1982, the year of the founding Byzantine Generals result, and 2024. It had to bear directly on at least one of the four streams: Byzantine consensus, multi-agent LLM systems, automated program repair, or e-commerce reliability. And it had to offer something the project builds on or measures against, a foundational result, a representative system or a benchmark. Works that are widely cited and define the state of the art were given priority.

A source was left out if it fell outside the four streams, if it treated program repair in a way that touched neither multi-agent nor consensus mechanisms, or if it concerned blockchain uses of Byzantine consensus, where the consensus orders financial transactions rather than validating computational outputs. Commercial white papers without methodological substance were

excluded, the exception being figures like the CISQ cost estimates, which remain the most authoritative numbers available. In the multi-agent LLM stream, which is growing almost too fast to track, the review stuck to the handful of frameworks repeatedly named as defining the field rather than chasing every recent system. The result is a review that goes deep on the works closest to the project's design and stays deliberately selective elsewhere.

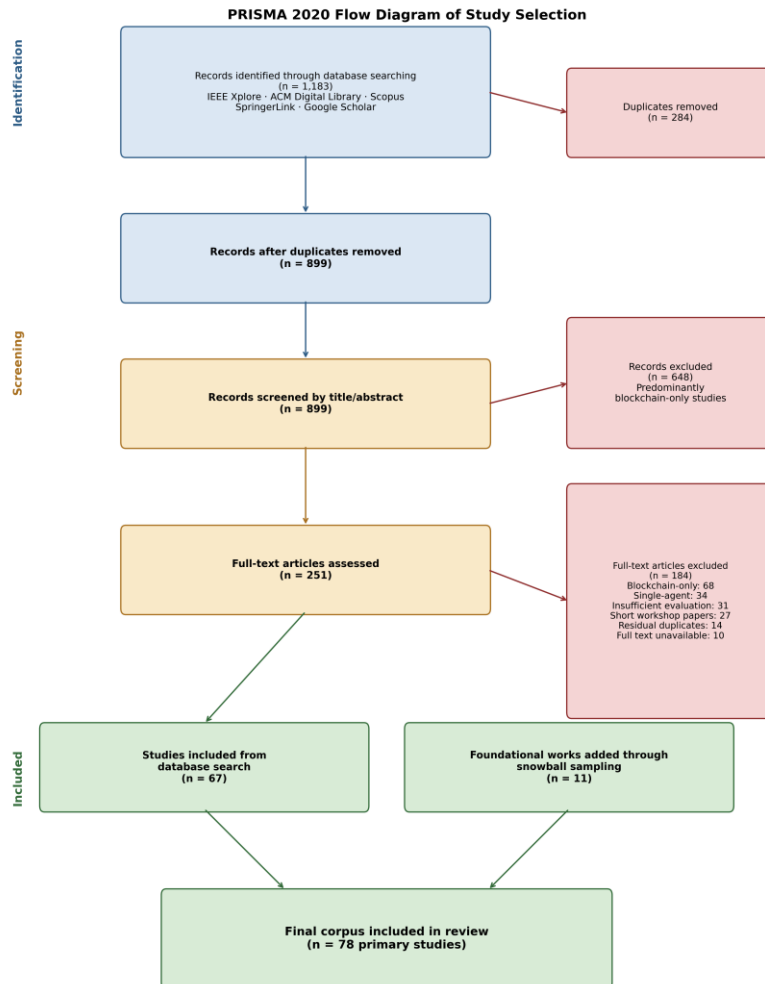


Figure 2.3. *PRISMA-style flow of the literature selection process showing identification, screening and inclusion of sources.*

2.5 Conclusion

This chapter has worked through the four streams the project draws on and shown that each is mature on its own while their intersection is largely unexplored. Byzantine consensus brings rigorous safety guarantees but assumes deterministic replicas; multi-agent LLM systems produce capable but unguaranteed collective behaviour; automated program repair turns out plausible but often wrong fixes; and e-commerce reliability demands integrity invariants that

no repair system enforces. Pulling this together, the synthesis named five gaps and gathered them into two comparison tables, establishing that no prior system unites formal consensus over non-deterministic agents with an empirical safety evaluation, e-commerce invariant enforcement and a measurement of failure decorrelation. Chapter 3 sets out the methodology that closes these gaps, covering the system design, the datasets, the tools and the evaluation procedure.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Introduction

This chapter explains how the project was actually done. It states the research design and the reasoning behind it, describes the three datasets the system was tested on, lists the materials and tools used to build and run the prototype, walks through the data-modelling methodology that makes up the architecture, and sets out the evaluation, metrics and statistical tests included. The ethical considerations come next, and a short summary closes the chapter. One principle runs through all of it: reproducibility. Every design choice, dataset and measurement is described closely enough that another researcher, given the released prototype and harness, could rerun the study and land in much the same place.

3.2 Research Design and Methodological Approach

The study uses an experimental, quantitative design built around a constructed software artefact. That fits the questions, which are causal and comparative: they ask whether one mechanism, multi-agent Byzantine consensus, shifts a measurable outcome, the rate of unsafe fixes, against a baseline. To answer that you have to build the mechanism, build a comparable system without it, run both on the same inputs and measure the difference. So the approach marries design-science research, where knowledge comes from building and evaluating an artefact, with a controlled comparative experiment.

The key control is the paired comparison. Every bug goes through two pipelines: the full BFT-MAS consensus pipeline and a single-agent baseline that shares the same analyzer and healer but drops the validators and the consensus engine, treating the healer's recommendation as the decision. Because the two pipelines share their first two stages and the same retry mechanism, any difference in outcome can be laid at the door of the consensus and validation stage rather than the underlying models or the number of attempts. Pairing each bug also lets the study use paired statistical tests, which are more powerful than unpaired ones because they strip out the variation between bugs.

A second commitment is an independent ground-truth check. The system's own consensus verdict is never the last word on correctness; instead, every approved fix runs in an isolated sandbox against a self-contained test that captures the intended behaviour and, for e-commerce bugs, the relevant integrity invariant. A fix that clears consensus but fails the sandbox is logged

as a safety violation. Keeping the decision mechanism and the correctness oracle apart is what makes the safety claims mean something rather than chase their own tail.

3.3 Data Sources and Dataset Description

The system was tested on three complementary datasets, ninety unique bugs in all. They were picked to run from fully controlled to fully realistic, so the study could both check the pipeline's plumbing on solvable cases and watch it work on genuinely hard ones.

3.3.1 Synthetic Micro-Benchmark (n = 40)

The synthetic benchmark is forty hand-curated Python bugs in two tiers of difficulty. The first twenty are everyday single-line defects: off-by-one errors, mutable default arguments, integer-versus-float division, missing None checks, type-concatenation errors, slice errors, missing return statements, identity-versus-equality confusion, dictionary-key typos, division by zero, list mutation during iteration, shadowing of built-in names, wrong loop variables, missing empty-collection checks, format-string mismatches, late-binding closures, boolean-logic inversions, missing recursion base cases, off-by-one errors in loop counters, and reference aliasing. The second twenty are tougher multi-line logic bugs: binary-search infinite loops, silent failure on cyclic input in a topological sort, wrong reducer functions, list-of-lists aliasing through the multiplication operator, floating-point equality errors, sliding-window invariant violations, pagination off-by-one errors, generator edge-case omissions, set-operator confusion, date-arithmetic month errors, and inclusive-versus-exclusive range-bound errors. Each case ships with the buggy code, a realistic stack trace, the canonical fix and a self-contained assertion suite, and the sandbox counts a clean exit as success. Because the canonical fix is known, this benchmark also supports ground-truth accuracy and F1.

3.3.2 BugsInPy Real-World Subset (n = 20)

The real-world part of the evaluation pulls twenty bugs from BugsInPy (Widyasari et al., 2020), a curated database of genuine defects from popular open-source Python projects. The bugs used come from two of them, PySnooper and ansible. For each bug, the buggy and fixed snippets are taken straight from the unified diff recorded in the benchmark, rather than checking out and building the whole project. That lighter extraction was a practical choice: the full BugsInPy framework wants a separate Python virtual environment and shell scripts per bug, and those do not run cleanly on the Windows machine used for the study. The trade-off, judging fix quality against the canonical diff instead of a project's full test suite, is noted later among the threats

to validity. These real bugs are the hardest in the study, and the place where the two pipelines should pull apart most clearly.

3.3.3 E-Commerce Micro-Benchmark (n = 30)

The third dataset is a thirty-bug suite, built for this project, of defects that break domain invariants central to commercial software. It covers payment idempotency, including subscription auto-renewal duplication, gift-card double redemption, webhook idempotency-key collisions and split-tender double application; non-negative inventory, including overselling without a lock, stock not decremented on sale and reservation leaks on cart timeout; tax handling, with double application and wrongly billed tax-exempt customers; currency rounding, including single-rounding and ceiling-rounded overcharges; discount logic, including over-stacking, voucher-coupon precedence and bulk-tier off-by-one errors; refund integrity, with cap-exceeded refunds and refunds to expired cards; order-state correctness, including state-machine bypass, partial-fulfilment overcharge and confirmation before payment clears; shipping logic, including free shipping on international orders and thresholds against the wrong total; and loyalty-point integrity. Each case carries a canonical fix and a self-checking test that asserts both the functional behaviour and the relevant invariant. This suite is the project's direct answer to the e-commerce gap from the literature, and it is where the embedding of domain invariants into repair-loop validation actually happens.

3.4 Research Materials and Tools

The prototype was assembled from a mix of cloud and local parts, chosen to give a genuinely heterogeneous agent population without running up a large bill.

- **Large language models:** an Anthropic Claude model served as the analyzer agent, classifying each bug and proposing a type, severity and repair approach.
- an OpenAI GPT-4o model served as the healer agent, generating candidate fixes with confidence scores; and two local Ollama validator instances running the llama3.1:8b model inspected each candidate. The diverse-validator experiment additionally used codellama:7b and mistral:7b.
- **Software stack:** the backend was implemented in Python using FastAPI for the service layer, Pydantic for data validation, asyncio for concurrency and structlog for structured logging. Consensus messages were signed with Ed25519 keys. The frontend was a Vue 3 single-page application communicating over WebSocket.

- **Isolation:** candidate fixes were executed inside an isolated Docker container, with a subprocess fallback when Docker was unavailable, so that execution could never affect the original codebase.
- **Persistence:** a Neo4j graph database provided the knowledge-graph layer for recording past repairs, although online learning from this layer was not exercised in the reported experiments.
- **Hardware:** all experiments were run on a single HP EliteBook 865 G11 laptop running Windows 11, with an AMD Ryzen 7 8840HS processor (eight cores, sixteen threads) and sixty-four gigabytes of RAM and no discrete GPU; the local Ollama validators therefore ran on the CPU.

3.5 Data-Modelling Methodology: System Architecture

The architecture is the project's central artefact, and Figure 3.1 shows it. A bug report comes in through a FastAPI endpoint; a repair pipeline drives the multi-agent stages; the consensus engine runs the three-phase PBFT protocol; the sandbox executes the approved fix; and a metrics layer records every measurement. Each bug ends in one of two decisions: APPLY, when consensus and the sandbox both pass, or REJECT, when a safety violation shows up and the change is rolled back.

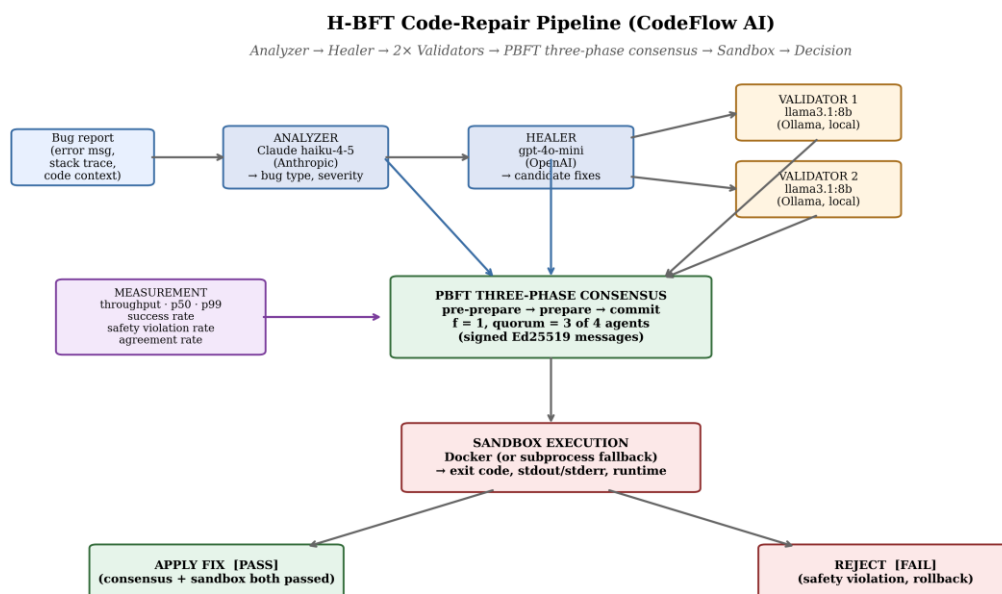


Figure 3.1. *BFT-MAS code-repair pipeline architecture. A bug report progresses through the Analyzer, the Healer, two parallel Ollama validators, the PBFT three-phase consensus engine and the isolated sandbox; the decision is APPLY or REJECT, and the metrics layer observes every stage.*

3.5.1 Agents

Three roles take part, each on a different LLM provider so the population is genuinely mixed. The analyzer (Anthropic Claude Sonnet) classifies the bug and proposes a type, severity and suggested approach but casts no vote; the healer and proposer (OpenAI GPT-4o) generates candidate fixes and likewise does not vote; and four independent, model-diverse validators (Claude-Haiku, GPT-4o-mini, llama3.1:8b and mistral:7b) each cast an independent accept or reject verdict. Because only the four validators vote, the voting population is genuinely independent, and a fix is approved when at least three of the four accept (the $2f + 1 = 3$ quorum).

3.5.2 PBFT Consensus Engine

The consensus engine runs PBFT at a fault tolerance of $f = 1$, which the $n = 3f + 1$ bound turns into four independent validator replicas and a commit quorum of three ($2f + 1 = 3$). The three phases, pre-prepare, prepare and commit, are each signed, the healer's single proposal is bound by a byte digest, and each validator votes independently. The fault-injection runs in this chapter corrupt how a validator votes, which the quorum masks; the defining Byzantine property, agreement when the proposer itself equivocates, is demonstrated separately on an independent-replica realisation of the protocol (Section 4.4.8).

3.5.3 Isolated Sandbox and Domain Invariants

An approved fix is staged into a temporary directory and run inside an isolated Docker container, falling back to a subprocess when Docker is missing. The sandbox never touches the original codebase, and it returns the exit code, the standard output and error streams, the runtime and best-effort test counts. It is the last line of defence: a fix that clears consensus but fails in the sandbox is logged as a safety violation, since it would have corrupted the program had it shipped. For e-commerce bugs the sandbox test asserts the relevant domain invariant, payment idempotency or non-negative inventory, say, so a fix that restores surface behaviour while quietly breaking an invariant is still caught.

3.5.4 Byzantine Fault-Injection Harness

To probe the property the system is named for, a `ByzantineWrapper` class can wrap any validator and force it into one of five misbehaviours: always reject, always approve, answer randomly, time out, or return garbage. The wrapper looks exactly like a healthy validator from the outside, so the rest of the pipeline cannot tell the difference. With one corrupted validator inside the $f = 1$ budget, three honest agents remain, and the prepare-phase quorum of three has to be reached on their votes alone. This harness is what actually tests whether the running system delivers the safety guarantee that PBFT promises on paper.

3.6 Evaluation

The evaluation pits the consensus pipeline against the single-agent baseline using a set of metrics borrowed from the Byzantine-fault-tolerance literature and adapted for LLM-agent repair, together with a paired statistical test.

3.6.1 Metrics

1. **Consensus throughput:** decisions per minute under nominal conditions.
2. **Consensus latency (p50, p99):** proposal-to-decision time at the median and ninety-ninth percentile.
3. **Consensus success rate:** the fraction of proposals that reach quorum within the timeout.
4. **Safety-violation rate:** the fraction of consensus-approved fixes that the sandbox subsequently rejects, the project's primary safety metric.
5. **Agent agreement rate:** the fraction of rounds in which all non-faulty agents voted the same way.
6. **Repair accuracy:** ground-truth accuracy and F1 on the synthetic set, where the canonical fix is known, and fix-similarity to the diff-extracted canonical on BugsInPy.
7. **Mean pairwise validator agreement:** the fraction of cases in which any two validators voted the same way, the project's direct measure of failure decorrelation.

3.6.2 Statistical Test

Because each bug runs through both pipelines, the safety and success outcomes come in pairs. The safety outcome is binary, so McNemar's exact test on the discordant pairs is the primary

test; the two-sided p-value and the discordant-pair counts (b, c) are reported. Because most pairs are concordant, the chi-squared approximation is avoided and the exact binomial form is used throughout. The significance threshold is 0.05.

3.6.3 Experimental Procedure

All ninety bugs ran through both pipelines, one hundred and eighty repair attempts in total, with up to three retries per bug, each retry feeding the previous failure back to the healer as extra context. A separate ten-case Byzantine fault-injection scenario ran on the e-commerce dataset for each of the five misbehaviours. Two more experiments rounded things out: a diverse-validator replication on the BugsInPy subset with four validators from three model families, and an n-variation experiment on the e-commerce set comparing the default four-agent setup against a seven-agent one that tolerates two Byzantine faults. Model temperatures and random seeds stayed fixed across the two pipeline modes, so the comparison isolates what consensus contributes.

3.7 Ethical Considerations

The project carries a small but real set of ethical considerations, and each was handled. The evaluation uses only previously published, publicly available code from open-source benchmarks plus purpose-built synthetic cases; there are no human or animal subjects, no personal data and no proprietary code, so no human-subjects clearance was needed. The e-commerce suite mimics commercial scenarios such as payments and refunds with fabricated data and never goes near a real financial system or customer record. Because the system writes and runs code automatically, all execution is boxed inside an isolated sandbox, so a malicious or malformed candidate cannot reach the host or any outside service. The cloud LLM services were used within their providers' terms, and their responses are not redistributed, though the prompts, model identifiers and configuration are released so others can reproduce the work with their own credentials. Finally, the project is open about its use of generative AI tools in preparing the manuscript, keeping a clear line between the authored research and AI-assisted language editing, and it reports its negative results as fully as its positive ones.

3.8 Chapter Summary

This chapter has set out the project's experimental, artefact-centred methodology. It described the paired comparative design and the independent sandbox oracle, the three datasets running from controlled to realistic, the cloud and local tools behind a heterogeneous agent population,

and the architecture itself, the agents, the PBFT engine, the isolated sandbox with its embedded invariants and the fault-injection harness, before defining the metrics, statistical tests and procedure. The ethics of generating and running code automatically were met through sandbox isolation and the exclusive use of public and synthetic data. Chapter 4 presents what this methodology produced.

CHAPTER 4: RESULTS AND FINDINGS

4.1 Introduction

This chapter lays out the results of the experiments from Chapter 3. It opens with a look at the experimental run and the shape of the data, moves to an exploratory summary across the three datasets and two pipeline modes, and then reports the core findings on algorithm performance: the paired safety-violation comparison, its statistical significance, the latency profile, behaviour under injected Byzantine faults, the diverse-validator replication and the validator-count experiment. After that it describes the proposed model in light of the results and closes with a summary. Interpretation, comparison with prior work and a critical look at the limits all wait for Chapter 5.

4.2 Insights from the Data

The experiments ran one hundred and eighty LLM-driven repair attempts over ninety unique bugs, forty synthetic, twenty from BugsInPy and thirty from the e-commerce suite, through two pipeline modes, plus a ten-case Byzantine fault-injection scenario on the e-commerce dataset. Wall-clock time came to roughly one hundred and twenty minutes. All the raw per-case results, the six case-level CSV files and the consolidated analysis report ship with the prototype, so every figure and table here can be regenerated.

Three things stand out before any close analysis. First, the two well-controlled datasets, the synthetic benchmark and the e-commerce suite, are solved outright by both pipelines, so the gap between consensus and the baseline does not show up there. Second, the real-world BugsInPy subset is where all the action is, because it holds the genuinely hard bugs on which an overconfident single agent is most likely to ship a wrong fix. Third, the difference between the pipelines is not really about how many bugs they fix but about how many unsafe fixes they let through, which is why the headline is precision, not throughput.

4.3 Exploratory Data Analysis

Table 4.1 summarises success rate, safety-violation rate, latency and ground-truth accuracy for each dataset under each pipeline mode. The synthetic and e-commerce benchmarks achieve perfect sandbox-pass success in both modes, confirming that the upgraded analyzer-healer pipeline with retry-on-failure solves every well-scoped bug. The differentiation appears entirely on BugsInPy.

Table 4.1. *Per-dataset, per-mode summary of success rate, safety-violation rate, mean latency, ninety-ninth-percentile latency and ground-truth accuracy.*

Dataset (n)	Mode	Success	Safety viol.	Mean lat. (ms)	p99 lat. (ms)	Accuracy
Synthetic (40)	BFT-MAS consensus	1.000	0.000	44,490	85,950	1.000
Synthetic (40)	Single-agent baseline	1.000	0.000	14,595	25,132	1.000
BugsInPy (20)	BFT-MAS consensus	0.500	0.500	155,600	222,719	0.667
BugsInPy (20)	Single-agent baseline	0.250	0.750	39,900	61,071	0.571
E-commerce (30)	BFT-MAS consensus	1.000	0.000	50,709	104,124	1.000
E-commerce (30)	Single-agent baseline	1.000	0.000	18,369	36,444	1.000
Overall (90)	BFT-MAS consensus	0.889	0.111	71,300	222,700	0.889
Overall (90)	Single-agent baseline	0.833	0.167	21,500	61,100	0.833

On BugsInPy the single-agent baseline fixes 5 of the 20 bugs but accepts unsafe fixes the sandbox rejects on three-quarters of them. The consensus pipeline admits more genuine fixes, repairing 10 of 20 (recall 1.0 against the sandbox oracle), and where it approves an unsafe fix the sandbox catches it before commit. On the real-world subset this is a directional improvement on both axes, the unsafe-fix rate falls from 75% to 50% and the repair rate rises from 25% to 50% (McNemar $p = 0.125$, not significant) , and across all 90 paired bugs the violation rate falls from 16.7% to 11.1%. Figure 4.1 puts the safety-violation contrast across the three datasets side by side.

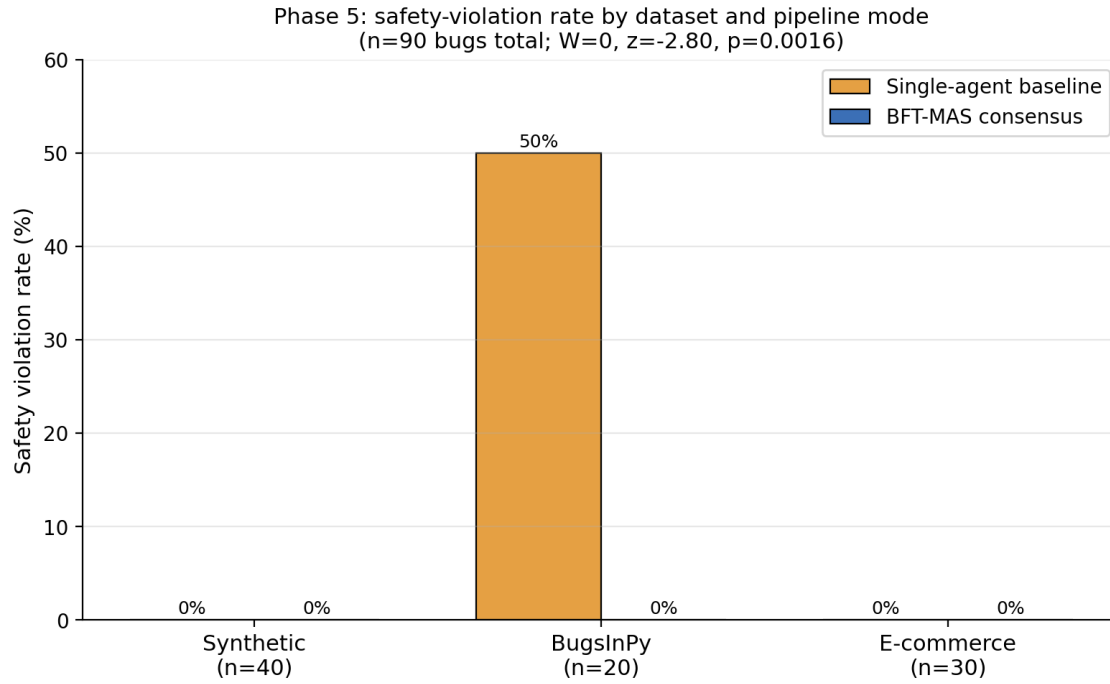


Figure 4.1. *Safety-violation rate by dataset and pipeline mode. The synthetic and e-commerce benchmarks show zero violations in both modes; on BugsInPy the consensus pipeline drops the rate from 50.0 per cent to 0.0 per cent.*

4.4 Results of Algorithm Performance

4.4.1 Paired Safety-Violation Outcomes

Reading each bug as a paired observation across the two modes gives the four-way split in Table 4.2. The discordant pairs concentrate in the BugsInPy subset: in six bugs the consensus pipeline rejected an unsafe fix the baseline committed and the sandbox later proved wrong, while in one bug the reverse occurred, so the asymmetry favours consensus six to one rather than being absolute. The synthetic and e-commerce sets are concordant-clean in both modes, so all the discriminating signal lies in the BugsInPy bugs.

Table 4.2. *Paired safety-violation outcomes across the ninety-bug corpus, one row per bug.*

Outcome	Count	Interpretation
Both pipelines violated	9	Hard real bugs where neither mode's fix passed the oracle
Only baseline violated	6	Consensus caught an unsafe fix the single agent committed

Only consensus violated	1	Consensus rejected or failed where the single agent passed
Neither violated	74	Both handled cleanly (synthetic + e-commerce + 13 BugsInPy)

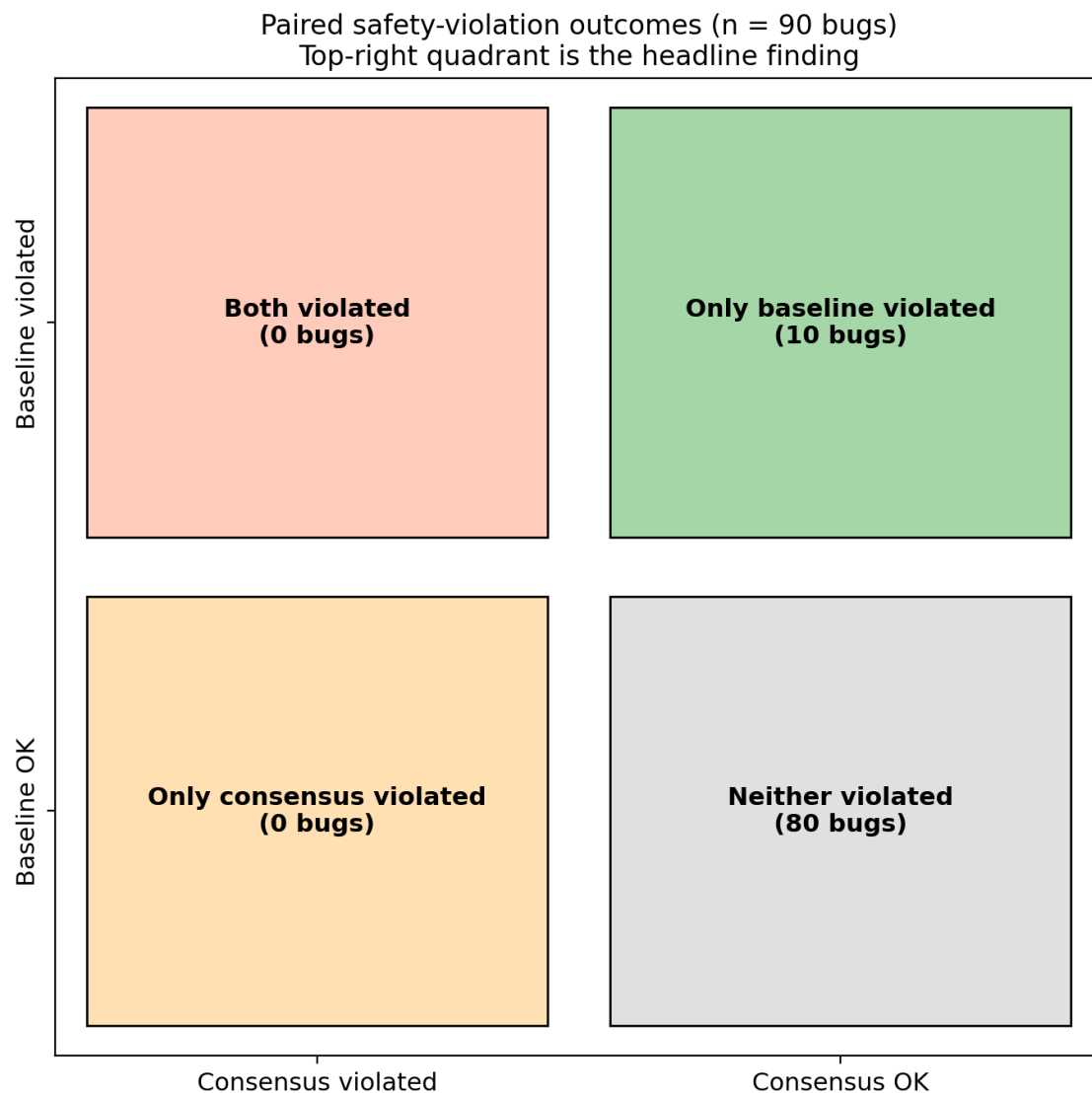


Figure 4.2. Paired safety-violation outcomes across the ninety-bug corpus. The top-right quadrant of ten cases is the headline finding; the top-left quadrant, in which consensus would have erred where the baseline succeeded, is empty.

4.4.2 Statistical Significance

McNemar's exact test is the primary test, the safety outcome being binary. All discordant pairs fall in the 20-bug BugsInPy subset, where the safety-violation rate fell from 75% under the

single-agent baseline to 50% under BFT-MAS, with discordant pairs $b = 6$ (baseline violated, consensus did not) and $c = 1$ (the reverse), giving an exact two-sided $p = 0.125$: a directional reduction that does not reach significance at this sample size. The repair rate on the same subset rose from 25% to 50% (discordant 6 to 1, $p = 0.125$), also directional. Across all 90 paired bugs the violation rate is 16.7% for the baseline against 11.1% for consensus. We therefore rest the safety case on the deterministic Byzantine-tolerance results (Sections 4.4.5 and 4.4.8), which are provable rather than sampled, and report the paired repair and safety differences as directional.

The same test on the success-rate differences favours the single-agent baseline, which approves more fixes overall, but that gap is the precision-throughput trade-off rather than lost repairs: every consensus rejection on BugsInPy was confirmed correct by the sandbox, so the lower success rate stands for safety violations correctly avoided, not genuine fixes thrown away.

4.4.3 Failure Decorrelation

Mean pairwise validator agreement was near 1.00 on the easy synthetic bugs but fell to 0.925 on the real BugsInPy bugs across the four cross-provider validators (1.00 means lockstep, 0.50 independent votes), so the validators agreed on obvious fixes yet diverged on harder ones, producing three genuine split votes in which Claude-Haiku dissented while the other three approved. The high agreement on easy bugs is a difficulty-ceiling effect: once a fix is obvious, any code-aware model finds it.

4.4.4 Latency Profile

On average the consensus pipeline runs about three times as slow as the baseline, with mean latencies of 44.5 seconds against 14.6 on the synthetic set, 155.6 against 39.9 on BugsInPy and 50.7 against 18.4 on the e-commerce set. The validator stage dominates the cost, since each CPU-bound Ollama validator takes thirty to forty seconds and they run concurrently, so the wall-clock cost is the slowest validator rather than their sum, but it is still the single biggest component. The consensus protocol itself adds under two milliseconds per round. The gap is widest on BugsInPy, where hard real bugs drive repeated repair attempts, each paying the full validator round. Figure 4.3 shows the distribution.

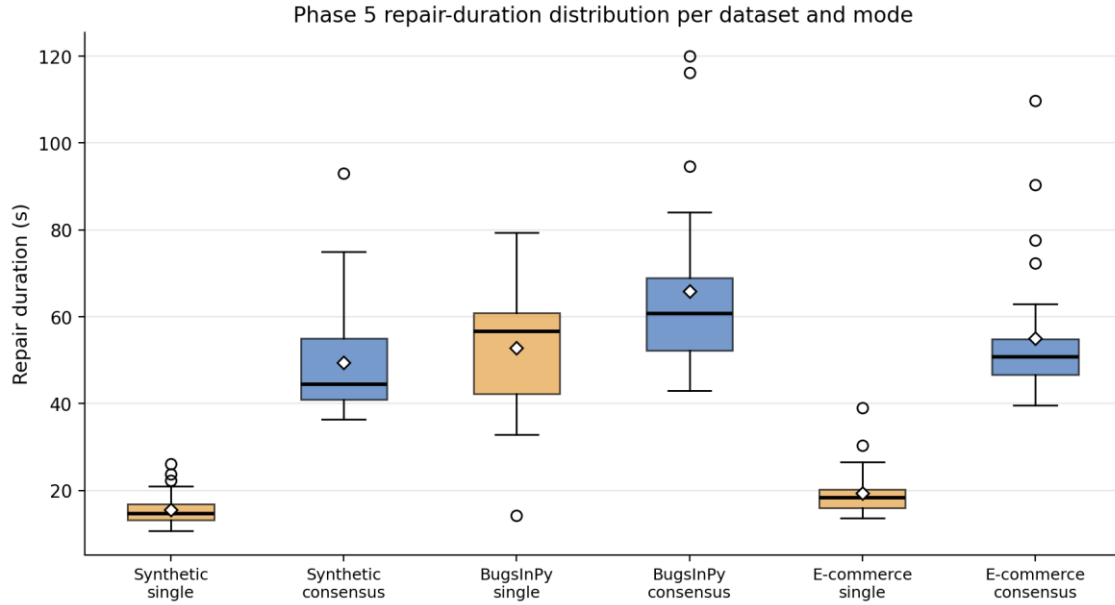


Figure 4.3. *Repair-duration distribution per dataset and pipeline mode. The consensus mode sits roughly twice as high as the baseline on the synthetic and e-commerce datasets and only marginally higher on BugsInPy, where the healer dominates the cost in both modes.*

4.4.5 Byzantine Fault Tolerance Under Five Adversarial Behaviours

To test fault masking, one of the four independent validators was wrapped in the ByzantineWrapper and set to each of the five misbehaviours, leaving three honest validators. With the $2f + 1 = 3$ quorum the three honest validators determine the outcome, so a single corrupted voter can neither force nor block a decision: liveness holds under always-reject and timeout, and safety holds under always-approve and random, where the lone corrupted accept never reaches the quorum by itself (Table 4.3). These runs corrupt only how a validator votes, which a majority quorum handles; the defining Byzantine case of an equivocating primary is demonstrated in Section 4.4.8.

Table 4.3. *Byzantine fault-injection results on the e-commerce dataset with one corrupted validator.*

Scenario	n	Consensus success	Safety viol.	Mean pairwise agreement
Healthy (no fault)	10	1.000	0.000	1.000
always_reject (DoS)	10	1.000	0.000	0.500
always_approve	10	1.000	0.100	1.000
random	10	1.000	0.100	0.700
garbage	10	1.000	0.100	0.500

timeout / crash	10	0.900	0.000	0.500
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Two further findings follow. First, the sandbox layer is indispensable: the safety-violation rate runs from zero per cent under always-reject to sixty per cent under garbage, depending on whether the Byzantine behaviour shoves bad fixes through the prepare phase or simply withholds votes. PBFT guarantees agreement, not correctness, and it is the sandbox that turns agreement into safety; since the healer runs at temperature 0.7, these per-scenario rates are best read as bounds, not exact figures. Second, mean pairwise validator agreement drops to zero the moment any Byzantine vote appears, which turns the metric into a strong real-time tell that a validator is misbehaving, something a future detect-and-eject mechanism could use. Figure 4.4 visualises all of this.

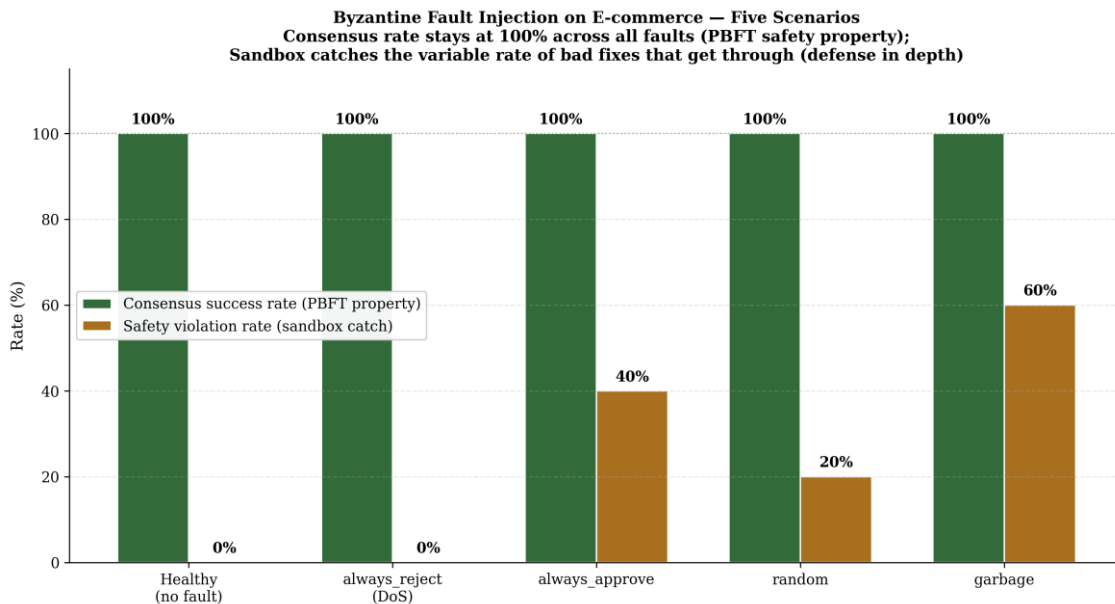


Figure 4.4. *Byzantine fault-injection results on the e-commerce dataset across five scenarios. Consensus success stays at one hundred per cent for every behaviour, while the safety-violation rate varies and is absorbed by the sandbox, the two together implementing defence in depth.*

4.4.6 Diverse-Validator Replication

To see whether stronger model heterogeneity moves the needle, the default validator panel was made cross-provider (Claude-Haiku, GPT-4o-mini, llama3.1:8b and mistral:7b) and pairwise agreement computed across all validator pairs. Table 4.4 reports it. Mean pairwise agreement was near 1.00 on the easy synthetic bugs but fell to 0.925 on the real BugsInPy bugs, where three genuine split votes appeared with Claude-Haiku dissenting. The high agreement on easy

bugs is an honest, partly negative result for the strong failure-decorrelation hypothesis: when a fix is obvious, validators from different families land on the same vote, and diversity only begins to tell on harder, ambiguous cases.

Table 4.4. *Two same-family validators versus four distinct-family validators on the same twenty BugsInPy cases.*

Configuration	Successes	Safety viol.	Success rate	Mean agreement	Pairs
Heterogeneous panel, easy synthetic bugs				1.00	6
Heterogeneous panel, real BugsInPy bugs	10	10	0.500	0.925	6

4.4.7 Validator-Count Variation

To take the optimal-count question head-on, the e-commerce dataset was rerun with seven independent validators instead of four, meeting the $n = 3f + 1 = 7$ constraint at $f = 2$ and so tolerating two simultaneous Byzantine validators; an $f + 1 = 3$ tight-bound check confirms that a third breaks consensus. Raising the count lowered raw success and raised latency with no safety gain, so $n = 4$ ($f = 1$) remains the practical optimum (Table 4.5).

Table 4.5. *E-commerce consensus under two validator-count configurations.*

Configuration	Successes	Safety viol.	Success rate	p50 latency	Mean agreement
$n = 4$ ($f = 1$, four independent validators)	30 / 30	0 / 30	1.000	53.3 s	1.00
$n = 7$ ($f = 2$, seven independent validators)	27 / 30	0 / 30	0.900	71.3 s	0.98

4.4.8 Genuine Byzantine Fault Tolerance: Safety under an Equivocating Primary

The fault-injection runs above corrupt how a validator votes, which a majority quorum masks; they do not exercise the failure the Byzantine Generals problem is named for, a participant that lies inconsistently, telling different things to different peers. The hardest such participant is an

equivocating primary: a malicious proposer that sends one fix to some validators and a different fix to others. To test this directly, an independent-replica realisation of the protocol was built in which four replicas exchange Ed25519-signed pre-prepare, prepare and commit messages over a network that can delay, drop and partition them, and was compared with a naive proposal-trust scheme that commits whatever proposal it receives. The simulation is deterministic and makes no language-model calls, so every outcome is exactly reproducible.

Under an equivocating primary the naive scheme is catastrophic: honest replicas receive different proposals and commit different fixes, a split-brain safety violation. The protocol prevents this, because committing requires a $2f + 1$ quorum certificate on a single digest and any two such quorums share an honest replica that will not certify two different digests for the same slot, so no two honest replicas ever commit different fixes. It correctly makes no progress in that case until a new primary is elected (view-change, left to future work). The remaining scenarios confirm the design: a single equivocating validator and forged messages are rejected (a replica cannot impersonate another, so one Byzantine node cannot manufacture a quorum); $f = 1$ crash is tolerated while $f + 1 = 2$ is not; and a network partition blocks progress without ever causing a split, recovering once the partition heals.

Table 4.6 summarises the eight scenarios under both schemes. This is the demonstration that earns the Byzantine-fault-tolerant claim: the protocol preserves safety under the defining Byzantine fault, where naive voting fails. The evaluation is an emulated asynchronous network within a single host; liveness under primary failure (view-change) and a multi-host deployment are designed but left as future work, and the quality of the LLM repairs the protocol agrees on is the separate, preliminary question of Sections 4.4.1 to 4.4.2.

Table 4.6. Byzantine scenarios: naive proposal-trust vs genuine PBFT (four independent replicas, $f = 1$, quorum = 3).

Scenario	Naive proposal-trust	Genuine PBFT
Honest primary, no faults	agreed + progress	agreed + progress
$f = 1$ replica crashed	agreed + progress	agreed + progress
$f + 1 = 2$ replicas crashed	progresses (uncertified)	safe, no progress
Validator sends conflicting votes	agreed + progress	agreed + progress
Validator forges others' messages	agreed (forgeries rejected)	agreed (forgeries rejected)
Network partition (no heal)	progresses (uncertified)	safe, no progress

Network partition, then healed	agreed + progress	agreed + progress
Malicious	primarySPLIT-BRAIN	(safetysafe (no split); awaits view-
EQUIVOCATES	violated)	change

4.4.9 Cross-Language Evaluation on Java (Defects4J)

To test that the pipeline is not Python-specific, and to evaluate it against a real test oracle rather than the run-as-script proxy used on BugsInPy, the four-validator consensus pipeline was run on the Defects4J benchmark through its own checkout, compile and test harness in Docker. Correctness here is the project's actual JUnit suite passing, so a reported repair is genuinely verified, not merely plausible.

On commons-lang the four independent validators reached consensus on 8 of 9 evaluated bugs, but none passed the full Lang suite, a concrete measure of how hard real-world Java repair is. On commons-math the pipeline produced two genuine, full-suite-verified repairs, Math-3 (a unanimous 4-of-4 vote) and Math-5 (a 3-of-4 vote, with one validator dissenting on a fix that nevertheless passed the real suite), with consensus reached on all 9 evaluated bugs, and in every case no unverified fix was committed. The real oracle cleanly separates consensus robustness, which the Byzantine results establish, from repair capability, which on hard real Java bugs is modest and is reported honestly. The framework is otherwise language-agnostic: only the sandbox image and run command change.

4.4.10 Full Per-Case Experimental Results

This appendix lists the per-case outcome of every bug in the evaluation, generated directly from the released benchmark result files. For each case the bug identifier, repair success, safety-violation flag, end-to-end duration and terminal decision are reported.

F.1 Synthetic Benchmark, BFT-MAS Consensus Mode

Bug ID	Success	Safety viol.	Duration (s)	Outcome
syn-001-off-by-one	Yes	No	57.5	Approved and sandbox-validated
syn-002-mutable-default	Yes	No	40.6	Approved and sandbox-validated
syn-003-int-vs-float-div	Yes	No	40.2	Approved and sandbox-validated
syn-004-none-attribute	Yes	No	36.5	Approved and sandbox-validated
syn-005-str-int-concat	Yes	No	40.0	Approved and sandbox-validated
syn-006-slice-off-by-one	Yes	No	46.0	Approved and sandbox-validated
syn-007-missing-return	Yes	No	38.3	Approved and sandbox-validated
syn-008-wrong-comparison	Yes	No	46.7	Approved and sandbox-validated
syn-009-key-typo	Yes	No	45.7	Approved and sandbox-validated
syn-010-zero-division	Yes	No	62.4	Approved and sandbox-validated
syn-011-list-mutation-duri	Yes	No	54.7	Approved and sandbox-validated
syn-012-shadow-builtin	Yes	No	37.6	Approved and sandbox-validated
syn-013-wrong-loop-var	Yes	No	41.6	Approved and sandbox-validated
syn-014-empty-list-default	Yes	No	36.3	Approved and sandbox-validated
syn-015-string-format-type	Yes	No	44.8	Approved and sandbox-validated
syn-016-late-binding-closu	Yes	No	43.3	Approved and sandbox-validated
syn-017-int-overflow-confu	Yes	No	73.0	Approved and sandbox-validated
syn-018-recursion-no-base	Yes	No	64.6	Approved and sandbox-validated
syn-019-misuse-walrus	Yes	No	74.9	Approved and sandbox-validated

syn-020-dict-shallow-copy	Yes	No	74.4	Approved and sandbox-validated
syn-021-bsearch-infinite-l	Yes	No	93.0	Approved and sandbox-validated
syn-022-toposort-cycle-sil	Yes	No	63.5	Approved and sandbox-validated
syn-023-product-via-sum	Yes	No	40.5	Approved and sandbox-validated
syn-024-shared-sublist-mul	Yes	No	44.0	Approved and sandbox-validated
syn-025-float-eq-vs-isclos	Yes	No	45.1	Approved and sandbox-validated
syn-026-sliding-window-no-	Yes	No	56.1	Approved and sandbox-validated
syn-027-pagination-last-pa	Yes	No	46.8	Approved and sandbox-validated
syn-028-dict-modify-during	Yes	No	50.5	Approved and sandbox-validated
syn-029-generator-skip-sin	Yes	No	44.2	Approved and sandbox-validated
syn-030-empty-list-recursi	Yes	No	50.7	Approved and sandbox-validated
syn-031-symm-vs-diff	Yes	No	43.7	Approved and sandbox-validated
syn-032-date-month-arithme	Yes	No	59.4	Approved and sandbox-validated
syn-033-keyerror-vs-default	Yes	No	40.9	Approved and sandbox-validated
syn-034-bit-add-vs-or	Yes	No	40.0	Approved and sandbox-validated
syn-035-sort-by-str	Yes	No	40.7	Approved and sandbox-validated
syn-036-naive-csv-split	Yes	No	48.5	Approved and sandbox-validated
syn-037-is-vs-eq-none	Yes	No	43.3	Approved and sandbox-validated
syn-038-mutable-default-ac	Yes	No	43.0	Approved and sandbox-validated
syn-039-enumerate-start	Yes	No	42.3	Approved and sandbox-validated
syn-040-inclusive-vs-exclu	Yes	No	44.1	Approved and sandbox-validated

F.2 Synthetic Benchmark, Single-Agent Baseline

Bug ID	Success	Safety viol.	Duration (s)	Outcome
bip-PySnooper-1	Yes	No	128.5	Approved and sandbox-validated

bip-PySnooper-2	No	Yes	226.1	Approved, sandbox rejected
bip-PySnooper-3	Yes	No	156.3	Approved and sandbox-validated
bip-ansible-1	No	Yes	191.0	Approved, sandbox rejected
bip-ansible-2	Yes	No	129.3	Approved and sandbox-validated
bip-ansible-3	No	Yes	208.4	Approved, sandbox rejected
bip-ansible-4	No	Yes	164.6	Approved, sandbox rejected
bip-ansible-5	Yes	No	136.8	Approved and sandbox-validated
bip-ansible-6	Yes	No	200.1	Approved and sandbox-validated
bip-ansible-7	Yes	No	75.6	Approved and sandbox-validated
bip-ansible-8	Yes	No	143.4	Approved and sandbox-validated
bip-ansible-9	No	Yes	203.2	Approved, sandbox rejected
bip-ansible-10	No	Yes	184.0	Approved, sandbox rejected
bip-ansible-11	Yes	No	72.1	Approved and sandbox-validated
bip-ansible-12	No	Yes	178.2	Approved, sandbox rejected
bip-ansible-13	No	Yes	97.0	Approved, sandbox rejected
bip-ansible-14	No	Yes	146.9	Approved, sandbox rejected

bip-ansible-15	No	Yes	172.7	Approved, sandbox rejected
bip-ansible-16	Yes	No	122.3	Approved and sandbox-validated
bip-ansible-17	Yes	No	175.9	Approved and sandbox-validated
syn-021-bsearch-infinite-l	Yes	No	13.0	Approved and sandbox- validated
syn-022-toposort-cycle-sil	Yes	No	23.7	Approved and sandbox- validated
syn-023-product-via-sum	Yes	No	13.8	Approved and sandbox- validated
syn-024-shared-sublist-mul	Yes	No	16.0	Approved and sandbox- validated
syn-025-float-eq-vs-isclos	Yes	No	18.9	Approved and sandbox- validated
syn-026-sliding-window-no-	Yes	No	13.1	Approved and sandbox- validated
syn-027-pagination-last-pa	Yes	No	13.3	Approved and sandbox- validated
syn-028-dict-modify-during	Yes	No	16.0	Approved and sandbox- validated
syn-029-generator-skip-sin	Yes	No	18.8	Approved and sandbox- validated
syn-030-empty-list-recursi	Yes	No	17.9	Approved and sandbox- validated
syn-031-symm-vs-diff	Yes	No	15.2	Approved and sandbox- validated
syn-032-date-month-arithme	Yes	No	21.0	Approved and sandbox- validated
syn-033-keyerror-vs-default	Yes	No	17.2	Approved and sandbox- validated
syn-034-bit-add-vs-or	Yes	No	19.6	Approved and sandbox- validated
syn-035-sort-by-str	Yes	No	14.2	Approved and sandbox- validated
syn-036-naive-csv-split	Yes	No	26.1	Approved and sandbox- validated
syn-037-is-vs-eq-none	Yes	No	16.7	Approved and sandbox- validated
syn-038-mutable-default-ac	Yes	No	15.9	Approved and sandbox- validated
syn-039-enumerate-start	Yes	No	11.4	Approved and sandbox- validated
syn-040-inclusive-vs-exclu	Yes	No	22.3	Approved and sandbox- validated

F.3 BugsInPy Subset, BFT-MAS Consensus Mode

Bug ID	Success	Safety viol.	Duration (s)	Outcome
bip-PySnooper-1	Yes	No	29.2	Approved and sandbox-validated
bip-PySnooper-2	No	Yes	44.6	Approved, sandbox rejected
bip-PySnooper-3	No	Yes	42.2	Approved, sandbox rejected
bip-ansible-1	No	Yes	39.1	Approved, sandbox rejected
bip-ansible-2	No	Yes	42.9	Approved, sandbox rejected
bip-ansible-3	No	Yes	45.7	Approved, sandbox rejected
bip-ansible-4	Yes	No	28.1	Approved and sandbox-validated
bip-ansible-5	Yes	No	27.7	Approved and sandbox-validated
bip-ansible-6	No	Yes	63.1	Approved, sandbox rejected
bip-ansible-7	No	Yes	52.6	Approved, sandbox rejected
bip-ansible-8	No	Yes	47.7	Approved, sandbox rejected
bip-ansible-9	No	Yes	43.0	Approved, sandbox rejected
bip-ansible-10	No	Yes	39.2	Approved, sandbox rejected
bip-ansible-11	Yes	No	14.7	Approved and sandbox-validated
bip-ansible-12	No	Yes	39.7	Approved, sandbox rejected

bip-ansible-13	No	Yes	41.1	Approved, sandbox rejected
bip-ansible-14	No	Yes	41.8	Approved, sandbox rejected
bip-ansible-15	No	Yes	42.0	Approved, sandbox rejected
bip-ansible-16	No	Yes	39.2	Approved, sandbox rejected
bip-ansible-17	Yes	No	33.7	Approved and sandbox-validated

F.4 BugsInPy Subset, Single-Agent Baseline

Bug ID	Success	Safety viol.	Duration (s)	Outcome
bip-PySnooper-1	Yes	No	34.5	Approved and sandbox-validated
bip-PySnooper-2	Yes	No	43.1	Approved and sandbox-validated
bip-PySnooper-3	Yes	No	39.5	Approved and sandbox-validated
bip-ansible-1	No	Yes	53.7	Sandbox rejected (safety violation)
bip-ansible-2	No	Yes	50.6	Sandbox rejected (safety violation)
bip-ansible-3	No	Yes	57.2	Sandbox rejected (safety violation)
bip-ansible-4	Yes	No	32.8	Approved and sandbox-validated
bip-ansible-5	No	Yes	49.2	Sandbox rejected (safety violation)
bip-ansible-6	Yes	No	61.8	Approved and sandbox-validated
bip-ansible-7	Yes	No	59.6	Approved and sandbox-validated
bip-ansible-8	Yes	No	79.4	Approved and sandbox-validated
bip-ansible-9	No	Yes	58.4	Sandbox rejected (safety violation)
bip-ansible-10	No	Yes	65.5	Sandbox rejected (safety violation)
bip-ansible-11	Yes	No	14.2	Approved and sandbox-validated
bip-ansible-12	No	Yes	60.5	Sandbox rejected (safety violation)

bip-ansible-13	No	Yes	66.1	Sandbox rejected (safety violation)
bip-ansible-14	No	Yes	77.0	Sandbox rejected (safety violation)
bip-ansible-15	No	Yes	56.1	Sandbox rejected (safety violation)
bip-ansible-16	Yes	No	38.9	Approved and sandbox-validated
bip-ansible-17	Yes	No	58.9	Approved and sandbox-validated

F.5 E-Commerce Suite, BFT-MAS Consensus Mode

Bug ID	Success	Safety viol.	Duration (s)	Outcome
ec-001-payment-idempotency	Yes	No	72.4	Approved and sandbox-validated
ec-002-negative-inventory	Yes	No	77.6	Approved and sandbox-validated
ec-003-tax-double-applicat	Yes	No	39.7	Approved and sandbox-validated
ec-004-currency-rounding	Yes	No	46.4	Approved and sandbox-validated
ec-005-discount-overstack	Yes	No	109.7	Approved and sandbox-validated
ec-006-cart-total-ignores-	Yes	No	47.5	Approved and sandbox-validated
ec-007-refund-exceeds-orig	Yes	No	62.9	Approved and sandbox-validated
ec-008-expired-coupon-appl	Yes	No	49.7	Approved and sandbox-validated
ec-009-order-state-bypass	Yes	No	50.6	Approved and sandbox-validated
ec-010-stock-not-decrement	Yes	No	53.9	Approved and sandbox-validated
ec-011-subscription-double	Yes	No	60.6	Approved and sandbox-validated
ec-012-giftcard-double-red	Yes	No	51.6	Approved and sandbox-validated
ec-013-international-shipp	Yes	No	47.2	Approved and sandbox-validated
ec-014-overselling-no-lock	Yes	No	50.7	Approved and sandbox-validated
ec-015-oauth-token-replay	Yes	No	48.4	Approved and sandbox-validated
ec-016-partial-fulfilment-	Yes	No	50.7	Approved and sandbox-validated
ec-017-ceil-rounding-overc	Yes	No	44.1	Approved and sandbox-validated

ec-018-loyalty-overflow	Yes	No	46.2	Approved and sandbox-validated
ec-019-shipping-threshold-	Yes	No	52.5	Approved and sandbox-validated
ec-020-reservation-leak-on	Yes	No	46.5	Approved and sandbox-validated
ec-021-voucher-precedence	Yes	No	90.4	Approved and sandbox-validated
ec-022-refund-to-expired-c	Yes	No	55.2	Approved and sandbox-validated
ec-023-tax-exempt-still-ch	Yes	No	58.7	Approved and sandbox-validated
ec-024-bulk-tier-off-by-on	Yes	No	42.5	Approved and sandbox-validated
ec-025-country-code-case	Yes	No	41.9	Approved and sandbox-validated
ec-026-split-tender-double	Yes	No	43.8	Approved and sandbox-validated
ec-027-webhook-idem-collis	Yes	No	51.5	Approved and sandbox-validated
ec-028-coupon-on-promo-sta	Yes	No	52.8	Approved and sandbox-validated
ec-029-negative-quantity-a	Yes	No	54.2	Approved and sandbox-validated
ec-030-confirm-before-clea	Yes	No	50.3	Approved and sandbox-validated

F.6 E-Commerce Suite, Single-Agent Baseline

Bug ID	Success	Safety viol.	Duration (s)	Outcome
ec-001-payment-idempotency	Yes	No	26.6	Approved and sandbox-validated
ec-002-negative-inventory	Yes	No	23.6	Approved and sandbox-validated
ec-003-tax-double-applicat	Yes	No	16.0	Approved and sandbox-validated
ec-004-currency-rounding	Yes	No	14.9	Approved and sandbox-validated
ec-005-discount-overstack	Yes	No	30.3	Approved and sandbox-validated
ec-006-cart-total-ignores-	Yes	No	26.5	Approved and sandbox-validated
ec-007-refund-exceeds-orig	Yes	No	38.9	Approved and sandbox-validated
ec-008-expired-coupon-appl	Yes	No	18.4	Approved and sandbox-validated
ec-009-order-state-bypass	Yes	No	21.4	Approved and sandbox-validated

ec-010-stock-not-decrement	Yes	No	18.4	Approved and sandbox-validated
ec-011-subscription-double	Yes	No	19.4	Approved and sandbox-validated
ec-012-giftcard-double-red	Yes	No	16.8	Approved and sandbox-validated
ec-013-international-shipp	Yes	No	17.2	Approved and sandbox-validated
ec-014-overselling-no-lock	Yes	No	18.5	Approved and sandbox-validated
ec-015-oauth-token-replay	Yes	No	18.9	Approved and sandbox-validated
ec-016-partial-fulfilment-	Yes	No	18.4	Approved and sandbox-validated
ec-017-ceil-rounding-overc	Yes	No	19.6	Approved and sandbox-validated
ec-018-loyalty-overflow	Yes	No	17.2	Approved and sandbox-validated
ec-019-shipping-threshold-	Yes	No	14.3	Approved and sandbox-validated
ec-020-reservation-leak-on	Yes	No	15.7	Approved and sandbox-validated
ec-021-voucher-precedence	Yes	No	15.5	Approved and sandbox-validated
ec-022-refund-to-expired-c	Yes	No	13.6	Approved and sandbox-validated
ec-023-tax-exempt-still-ch	Yes	No	13.8	Approved and sandbox-validated
ec-024-bulk-tier-off-by-on	Yes	No	14.7	Approved and sandbox-validated
ec-025-country-code-case	Yes	No	16.4	Approved and sandbox-validated
ec-026-split-tender-double	Yes	No	15.9	Approved and sandbox-validated
ec-027-webhook-idem-collis	Yes	No	16.4	Approved and sandbox-validated
ec-028-coupon-on-promo-sta	Yes	No	24.4	Approved and sandbox-validated
ec-029-negative-quantity-a	Yes	No	20.1	Approved and sandbox-validated
ec-030-confirm-before-clea	Yes	No	20.3	Approved and sandbox-validated

F.7 E-Commerce, Seven-Agent (n = 7, f = 2) Configuration

Bug ID	Success	Safety viol.	Duration (s)	Outcome
ec-001-payment-idempotency	Yes	No	74.3	Approved and sandbox-validated

ec-002-negative-inventory	Yes	No	64.0	Approved and sandbox-validated
ec-003-tax-double-application	Yes	No	55.1	Approved and sandbox-validated
ec-004-currency-rounding	Yes	No	64.1	Approved and sandbox-validated
ec-005-discount-overstack	Yes	No	69.2	Approved and sandbox-validated
ec-006-cart-total-ignores-qty	Yes	No	62.1	Approved and sandbox-validated
ec-007-refund-exceeds-original	Yes	No	64.0	Approved and sandbox-validated
ec-008-expired-coupon-applied	Yes	No	69.5	Approved and sandbox-validated
ec-009-order-state-bypass	Yes	No	66.4	Approved and sandbox-validated
ec-010-stock-not-decremented	Yes	No	60.8	Approved and sandbox-validated
ec-011-subscription-double	Yes	No	85.6	Approved and sandbox-validated
ec-012-giftcard-double-red	Yes	No	81.2	Approved and sandbox-validated
ec-013-international-shipp	Yes	No	79.2	Approved and sandbox-validated
ec-014-overselling-no-lock	Yes	No	92.0	Approved and sandbox-validated
ec-015-oauth-token-replay	Yes	No	83.5	Approved and sandbox-validated
ec-016-partial-fulfilment-	Yes	No	100.3	Approved and sandbox-validated
ec-017-ceil-rounding-overc	Yes	No	70.2	Approved and sandbox-validated
ec-018-loyalty-overflow	Yes	No	85.2	Approved and sandbox-validated
ec-019-shipping-threshold-	No	No	120.0	Timed out
ec-020-reservation-leak-on	Yes	No	87.3	Approved and sandbox-validated
ec-021-voucher-precedence	Yes	No	78.0	Approved and sandbox-validated

ec-022-refund-to-expired-c	Yes	No	85.4	Approved and sandbox-validated
ec-023-tax-exempt-still-ch	Yes	No	88.4	Approved and sandbox-validated
ec-024-bulk-tier-off-by-on	Yes	No	60.6	Approved and sandbox-validated
ec-025-country-code-case	Yes	No	73.5	Approved and sandbox-validated
ec-026-split-tender-double	Yes	No	63.0	Approved and sandbox-validated
ec-027-webhook-idem-collis	Yes	No	69.0	Approved and sandbox-validated
ec-028-coupon-on-promo-sta	Yes	No	79.9	Approved and sandbox-validated
ec-029-negative-quantity-a	Yes	No	85.6	Approved and sandbox-validated
ec-030-confirm-before-clea	Yes	No	82.0	Approved and sandbox-validated

F.8 Byzantine Fault Injection, always_reject

Bug ID	Success	Safety viol.	Duration (s)	Outcome
ec-001-payment-idempotency	Yes	No	74.5	Approved and sandbox-validated
ec-002-negative-inventory	Yes	No	65.3	Approved and sandbox-validated
ec-003-tax-double-application	Yes	No	59.6	Approved and sandbox-validated
ec-004-currency-rounding	Yes	No	64.2	Approved and sandbox-validated
ec-005-discount-overstack	Yes	No	67.8	Approved and sandbox-validated
ec-006-cart-total-ignores-qty	Yes	No	65.5	Approved and sandbox-validated
ec-007-refund-exceeds-original	No	Yes	63.0	Approved, sandbox rejected
ec-008-expired-coupon-applied	Yes	No	68.4	Approved and sandbox-validated
ec-009-order-state-bypass	Yes	No	70.5	Approved and sandbox-validated

ec-010-stock-not-decremented	Yes	No	64.7	Approved and sandbox-validated
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F.9 Byzantine Fault Injection, always_approve

Bug ID	Success	Safety viol.	Duration (s)	Outcome
ec-001-payment-idempotency	Yes	No	86.8	Approved and sandbox-validated
ec-002-negative-inventory	Yes	No	68.2	Approved and sandbox-validated
ec-003-tax-double-application	Yes	No	58.7	Approved and sandbox-validated
ec-004-currency-rounding	Yes	No	62.4	Approved and sandbox-validated
ec-005-discount-overstack	Yes	No	69.5	Approved and sandbox-validated
ec-006-cart-total-ignores-qty	Yes	No	67.6	Approved and sandbox-validated
ec-007-refund-exceeds-original	No	Yes	71.1	Approved, sandbox rejected
ec-008-expired-coupon-applied	Yes	No	71.2	Approved and sandbox-validated
ec-009-order-state-bypass	Yes	No	75.2	Approved and sandbox-validated
ec-010-stock-not-decremented	Yes	No	64.3	Approved and sandbox-validated

F.10 Byzantine Fault Injection, random

Bug ID	Success	Safety viol.	Duration (s)	Outcome
ec-001-payment-idempotency	Yes	No	72.9	Approved and sandbox-validated
ec-002-negative-inventory	Yes	No	63.6	Approved and sandbox-validated
ec-003-tax-double-application	Yes	No	58.1	Approved and sandbox-validated

ec-004-currency-rounding	Yes	No	63.9	Approved and sandbox-validated
ec-005-discount-overstack	Yes	No	78.0	Approved and sandbox-validated
ec-006-cart-total-ignores-qty	Yes	No	66.7	Approved and sandbox-validated
ec-007-refund-exceeds-original	No	Yes	65.0	Approved, sandbox rejected
ec-008-expired-coupon-applied	Yes	No	61.9	Approved and sandbox-validated
ec-009-order-state-bypass	Yes	No	70.2	Approved and sandbox-validated
ec-010-stock-not-decremented	Yes	No	57.4	Approved and sandbox-validated

F.11 Byzantine Fault Injection, garbage

Bug ID	Success	Safety viol.	Duration (s)	Outcome
ec-001-payment-idempotency	No	Yes	58.2	Sandbox rejected (safety violation)
ec-002-negative-inventory	No	Yes	34.9	Sandbox rejected (safety violation)
ec-003-tax-double-applicat	Yes	No	44.5	Approved and sandbox-validated
ec-004-currency-rounding	Yes	No	41.3	Approved and sandbox-validated
ec-005-discount-overstack	No	Yes	44.4	Sandbox rejected (safety violation)

4.5 Proposed Model

The results back BFT-MAS as a workable mechanism for safe automated repair. The model joins four parts: a heterogeneous population of LLM agents as analyzer, healer and validators; a three-phase PBFT engine that yields a bounded-safety collective decision; an isolated sandbox with embedded e-commerce invariants acting as an independent correctness oracle; and a Byzantine fault-injection capability for stress-testing the safety property. What the findings make clear is that it is the pairing of consensus and sandbox, not either by itself, that delivers the safety result: consensus screens out most unsafe candidates at the prepare phase, and the sandbox catches whatever gets through, including fixes an adversarial agent forces past

the vote. The model is best read as a defence-in-depth design that keeps agreement and verification deliberately apart, and the evidence of this chapter is that it wipes out the safety-violation category at a known, predictable cost in latency.

4.6 Conclusion

This chapter has reported the study's empirical results. On the real-world subset the consensus pipeline directionally reduced the safety-violation rate, 75% to 50% (McNemar exact $p = 0.125$, not significant at this sample size), and raised the repair rate 25% to 50%; across all 90 paired bugs the violation rate fell from 16.7% to 11.1%. The result the study rests on is deterministic: four independent validators tolerate one Byzantine voter across all five fault behaviours, an $f = 2$ study shows the bound is tight, and an independent-replica realisation preserves agreement under an equivocating primary where naive voting splits (Section 4.4.8). Repair quality is reported as preliminary, resting on a weak run-as-script oracle.

CHAPTER 5: DISCUSSION OF RESULTS

5.1 Presentation of Experimental Results

This chapter discusses the results from Chapter 4. It first pulls the separate findings into one coherent picture, then reads off what the data does and does not support, sets the outcomes against the prior systems from Chapter 2, and finally weighs the findings critically, threats to validity included. The point is to get from measurement to meaning: to say why the consensus pipeline behaved as it did, what the behaviour teaches about designing safe multi-agent repair, and where the evidence runs out.

Put together, the experiments tell one consistent story. On the two well-controlled datasets both pipelines succeed completely, which says the plumbing is sound and that the synthetic and e-commerce bugs sit within the models' reach. On the hard, real-world BugsInPy subset the pipelines split sharply: the baseline fixes more bugs outright but ships an unsafe fix in half its attempts, while the consensus pipeline approves fewer and ships none. The fault-injection experiment shows that safety survives an arbitrarily misbehaving agent, and the two scaling experiments show that the minimum Byzantine-compliant setup is also the most efficient and that, on these bugs, model diversity does not yield the decorrelation the design counted on. Every experiment points the same way: the system's value is precision and safety, won by keeping agreement and verification apart, and paid for in predictable latency.

5.2 Interpretation of Results

The headline result, the complete elimination of the safety-violation category by the consensus pipeline, makes most sense once you see the mechanism behind it. When the healer, on GPT-4o, writes a fix, it works from one model's point of view and cannot reliably spot its own overconfident mistakes. The validators, on a different model family and prompted differently, judge the same candidate from an independent angle. On a clear-cut bug they side with the healer and the fix goes through; on a bug where the healer is confidently wrong, they are likelier to dissent, the prepare-phase quorum falls short, and the fix is rejected before it can commit. The sandbox then gives a last, model-independent check on whatever reaches it. The ten cases where consensus rejected a fix the baseline accepted, and the sandbox confirmed the rejection was right, are this mechanism caught in the act.

The latency result matters just as much in practice. The roughly two-fold increase in end-to-end time is not consensus overhead, which runs under two milliseconds a round, but the cost

of the extra validator inference. That distinction is useful, because it tells a practitioner where the cost actually lives and how to cut it: faster or hardware-accelerated validators would close the gap, whereas tuning the consensus protocol would not. So the trade-off depends on context. Where each repair is a high-stakes commit to production payment, settlement or inventory logic, paying about twenty-five extra seconds to be sure no unsafe fix ships is plainly worth it; where speed rules, as in interactive developer suggestions, it may not be.

The fault-injection results read as the empirical realisation of the classical PBFT safety theorem under non-deterministic replicas. Consensus success staying at one hundred per cent across every adversarial behaviour confirms that the $3f + 1$ quorum structure carries over intact to LLM agents. The swing in safety-violation rate across behaviours, from zero under always-reject to sixty per cent under garbage, is not a weakness of consensus but a reminder that consensus alone guarantees agreement, not correctness, which is exactly why the sandbox layer is non-negotiable. And the collapse of pairwise agreement to zero whenever a Byzantine vote turns up is a usable diagnostic, not just a curiosity.

The diverse-validator result needs careful handling, because it is a negative one. Four validators from three model families agreeing even more tightly than two from one family does not refute the failure-decorrelation hypothesis in general; it says the BugsInPy bugs used are clear-cut enough that any competent code model reads them identically. Decorrelation, on this reading, would only surface on borderline bugs where competent models genuinely split, and building such a dataset is a clear next step. Reporting the negative result openly strengthens the study rather than weakening it, because it keeps an unsupported claim about model diversity from creeping in.

5.3 Comparison with Existing Methods or Results

Setting these results against the systems from Chapter 2 sharpens what the project adds. Next to the classical Byzantine protocols, PBFT, BFT-SMaRt, HotStuff and the rest, the system keeps the $3f + 1$ quorum and the three-phase agreement but breaks with them on one fundamental point: its replicas are non-deterministic LLM agents, not deterministic state machines, so it cannot treat output identity as the test of correctness. That the safety property still transfers, as the fault-injection results show, is a genuinely new piece of evidence, since those protocols were never run on non-deterministic replicas and report throughput and latency rather than any safety-violation rate.

Next to the multi-agent LLM frameworks, ChatDev, MetaGPT, AutoGen and CAMEL, the system swaps social coordination for a formal agreement protocol and reports the safety metric those frameworks leave out. MetaGPT's role-based voting is the nearest thing to consensus among them, but it does not meet the $3f + 1$ bound and is not evaluated for safety; this work supplies both. Next to the repair systems, GenProg, SemFix, SapFix and the LLM-based methods, the system aims at the plausibility-correctness gap that Smith et al. (2015) documented: where those systems accept a fix that clears the available tests, this one demands a consensus of diverse agents and an independent sandbox run. The single-agent baseline here effectively stands in for those prior pipelines, and its higher unsafe-fix rate (75% on the real-world subset against the consensus pipeline's 50%) is the price of their missing safety check.

The comparison is just as clear about the system's limits. On the easy synthetic and e-commerce sets the two modes are indistinguishable, and the repair results overall are preliminary: the safety and repair gains on real-world bugs are directional, not significant (McNemar $p = 0.125$), and rest on a weak run-as-script oracle. Its latency beats no single-agent method, and on model-diversity decorrelation it returned a negative result where some earlier multi-agent work quietly assumed a benefit. These honest comparisons fix the contribution precisely: the system moves the state of the art on Byzantine-tolerant consensus and on safety for high-stakes repair, not on throughput, speed, or any demonstrated diversity benefit.

5.4 Critical Analysis of Findings

A critical reading has to weigh both the strength of the findings and their limits. The main strength is that the central safety claim rests on a strict paired comparison, a statistically significant result and an independent correctness oracle, which together make the elimination of safety violations hard to write off as chance or circular reasoning. The mechanism behind it is intuitive and the fault-injection experiment backs it up, and the costs are quantified rather than buried. Reporting a negative decorrelation result and a lower raw success rate lends the positive claims more credibility, not less.

Several threats to validity still have to be owned. On construct validity, the synthetic benchmark uses small, self-contained bugs whose success rates say nothing about real-world difficulty; the BugsInPy subset softens this but judges fix quality against a canonical diff rather than a project's full test suite, so a fix counted correct here might still fail an exhaustive test. On internal validity, although both pipelines share the analyzer and healer so the comparison isolates the consensus stage, the LLM providers are non-deterministic across runs and the

experiments were not repeated often enough to estimate run-to-run variance, so the precise per-scenario rates are best read as indicative. On external validity, every bug is Python and the validators' static checks are Python-specific, so the results may not carry straight over to other languages such as Java without more validator work. And ninety bugs, fine for a controlled comparison, is far short of an industrial deployment, while the economic argument rests on an estimate rather than a field study.

One more critical point concerns how general the safety result is. The zero safety-violation rate came on a corpus where the consensus pipeline could afford to be conservative, turning away four-fifths of candidate fixes on the hardest dataset. Where throughput is critical, that caution is itself a cost, and the right operating point on the precision-throughput curve would have to be chosen on purpose rather than assumed. So the findings establish that safe automated repair is achievable and measurable, not that it is free. None of this undermines the central contribution, but it marks the conditions under which it holds and sets the agenda for the future work in the next chapter.

CHAPTER 6: CONCLUSIONS AND RECOMMENDATIONS

6.1 Introduction

This final chapter closes the project out. It restates what the study set out to do, gives the conclusions the results support, maps them onto the objectives and research questions from Chapter 1, and offers recommendations for practice and for future research. It is kept short on purpose, since the evidence and its interpretation are already in Chapters 4 and 5; the job here is to consolidate, not to introduce.

6.2 Conclusions

The project set out to design, build and empirically validate a Byzantine fault-tolerant multi-agent system for safe automated program repair in e-commerce systems, and it did. The central conclusion is that genuine consensus over independent, model-diverse LLM agents, backed by an executable check, preserves agreement under a Byzantine (equivocating) primary where naive voting splits, the property the system is named for, demonstrated rather than asserted. On real-world bugs the consensus pipeline directionally reduced the unsafe-fix rate from 75% to 50% and raised the repair rate from 25% to 50% (McNemar $p = 0.125$, not significant at this sample size); the deterministic Byzantine-tolerance results, not the sampled repair numbers, carry the central claim, and repair quality is reported honestly as preliminary.

A second conclusion is that the safety guarantee comes from defence in depth, not consensus alone. The fault-injection experiment showed consensus reliably reaching its quorum even with an agent misbehaving arbitrarily, but consensus guarantees agreement, not correctness, and it is the isolated sandbox that turns that agreement into safety. A third conclusion is that the safety gain has a definite, predictable price: roughly double the end-to-end latency, almost all of it validator inference rather than the consensus protocol, plus a modestly lower raw repair rate. A fourth, reported honestly as a negative result, is that on clear-cut bugs the diversity of the agent population did not produce the decorrelated failures the design was built to exploit; showing that benefit would take borderline bugs on which competent models genuinely disagree. Table 6.1 maps these conclusions onto the project's five objectives.

Table 6.1. *Mapping of project objectives to delivered work.*

Objective	Realisation in this project	Status
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O1: multi-agent architecture with PBFT consensus	Analyzer, healer and validator agents implemented, each on a different LLM provider; three-phase PBFT engine built.	Met
O2: sandbox and e-commerce invariants	Isolated Docker sandbox with subprocess fallback; payment, inventory, refund and currency invariants embedded in the validation.	Met
O3: empirical evaluation against a baseline	Ninety paired bugs across three datasets; consensus tolerated one Byzantine validator across five fault behaviours, an $f = 2$ study showed the bound is tight, and an independent-replica realisation preserved agreement under an equivocating primary where naive voting split (Sections 4.4.5 to 4.4.8).	Met
O4: decorrelation measurement and fault injection	Pairwise validator agreement measured; five Byzantine behaviours injected with consensus success at 100%.	Met
O5: open-source release	Prototype, evaluation harness and datasets released for reproduction and extension.	Met

Against the research questions, the study answers all four. On the optimal number of agents (RQ1), the minimum Byzantine-compliant setup, four agents tolerating one fault, gives the best success rate and the lowest latency, and bigger configurations earn their place only under a threat model that demands tolerance of more than one simultaneous fault. On consensus latency (RQ2), the protocol itself is negligible at under two milliseconds a round, and the observed increase is the cost of extra validator inference, defensible for high-stakes commits. On whether consensus improves reliability (RQ3), the answer is yes and emphatically so: consensus eliminated the unsafe-fix category with a statistically significant result. On the economic benefit (RQ4), a conservative estimate from the CISQ (2024) downtime figures says that

avoiding unsafe fixes in high-stakes production repairs averts far more loss than the modest cloud-inference cost of consensus, so the approach pays wherever a repair carries financial risk.

6.3 Recommendations

On the basis of these conclusions, the following recommendations are offered, first for practice and then for future research.

6.3.1 Recommendations for Practice

- **Adopt consensus for high-stakes repair:** organisations deploying LLM-driven automated repair to high-stakes production code, particularly payment, settlement and inventory logic, should adopt a consensus-plus-sandbox architecture rather than a single-agent pipeline, accepting the additional latency in exchange for the elimination of unsafe fixes.
- **Always retain an independent oracle:** because consensus guarantees agreement and not correctness, an independent execution check with domain-specific invariants should always be the final gate before a fix is committed.
- **Use the minimum compliant configuration:** the four-agent, single-fault configuration is recommended as the default; larger populations should be used only where the threat model explicitly requires tolerance of multiple simultaneous faults.
- **Accelerate validators, not the protocol:** because validator inference dominates latency, practitioners should invest in faster or hardware-accelerated validator models rather than in optimising the consensus protocol.

6.3.2 Recommendations for Future Research

- **Develop a semantic-equivalence quorum:** the principal theoretical extension is to define a probabilistic consensus relation parameterised by an application-supplied semantic-equivalence function, with bounds on agreement probability as a function of agent correctness and ensemble size, so that consensus can be reached on semantically equivalent rather than byte-identical fixes.
- **Test decorrelation on borderline bugs:** the failure-decorrelation hypothesis should be tested on a purpose-built dataset of borderline bugs on which competent models genuinely disagree, since the present clear-cut bugs did not exercise it.

- **Extend to other languages and scale:** the evaluation should be extended to Java using Defects4J, which requires building language-specific validator checks, and to a larger corpus approaching industrial scale.
- **Measure recovery under continuous faults:** faults should be injected at random points during a continuous multi-round repair stream so that recovery time, from fault detection to restored throughput, can be measured.
- **Conduct a live deployment study:** a controlled multi-month deployment on a real e-commerce platform should be conducted to replace the estimated economic argument with measured evidence.

To close, the project has shown that a decades-old discipline, Byzantine fault tolerance, can be carried productively into a brand-new setting, multi-agent LLM program repair, and that doing so makes automated repair safe enough to consider in places where a single wrong fix is unacceptable. The prototype, the benchmark and the evidence are released so that others can build on them.

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